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POPULATION GENOMICS OF REDBREAST SUNFISH (*LEPOMIS AURITUS*): EXPLORING GENE FLOW AND LOCAL ADAPTATION IN A WIDELY DISTRIBUTED FRESHWATER FISH

by

GARRET STRICKLAND

(Under the Direction of James H. Roberts)

ABSTRACT

Little information is available concerning the distribution of genetic diversity in non-salmonid, non-imperiled, freshwater fish. In order to fill in this knowledge gap, I conducted a population genomics survey in Redbreast Sunfish (*Lepomis auritus*; RBS), a widespread, generalist species distributed along the Atlantic slope rivers of eastern North America. I sampled four basins (ACF, Savannah, Roanoke, and James) at eight sites each with a factorial experimental design. Sites were distributed among coastal plain, Piedmont, or mountain ecoregions in order to capture the greatest range of environmental states experienced by RBS, with the intention of finding evidence for local adaptation to these distinct environments. Individuals were genotyped via ddRADseq resulting in 2858 single nucleotide polymorphisms (SNPs), of which 287 were identified as putatively adaptive via outlier tests and genotype-environment association analyses. The 2571 putatively neutral SNPs allowed for delineating the scales of population structure and the influence of landscape factors on genetic diversity and differentiation. Overall, I found RBS to show hierarchical population structure whereby the majority (67.5%) of molecular variation occurred among basins with decreasing variation among ecoregions within basins (7.4%), and among sites within ecoregions (1.7%). Using linear mixed models, I also found that isolation by riverine distance (IBD) was supported in the two southern basins but not the two in the north, possibly due to the influence of Pleistocene glaciations on contemporary population structure. In addition, a small subset of pairwise comparisons allowed for testing the effect of ecoregion on IBD; this analysis showed that the IBD slope was steeper in the mountain and Piedmont ecoregions than in the coastal plain, suggesting less restricted gene flow in the coastal plain. In addition, random forest models suggest that major dams produce the same effect on genetic differentiation as approximately 450 riverine kilometers, whereas impoundments have little effect on differentiation. Genetic diversity did not vary consistently by ecoregion or stream size. However, a random forest model

did show higher genetic diversity in populations with warmer temperature means and higher cation exchange capacity, indicative of larger population sizes due to higher growth rates at higher temperatures and higher fitness in environments with more stable flow regimes and higher primary productivity. The putatively adaptive SNPs provided some evidence for local adaptation as 188 SNPs where significantly associated with physiographic or climatic variation. However, putatively adaptive differentiation was strongly correlated with neutral differentiation, suggesting the need for further "ground-truthing" of such putatively adaptive SNPs in this and other studies.

INDEX WORDS: Genomics, Freshwater fish, Centrarchidae, Lepomis, Population genetics, Landscape genetics, Stream ecology, Evolution

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GARRET STRICKLAND

B.S., Georgia Southern University, 2015

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Requirements for the Degree

MASTER OF SCIENCE

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CHAPTER 1: INTRODUCTION

Purpose of this Study

Redbreast Sunfish (*Lepomis auritus*; hereafter RBS), a widely distributed freshwater fish occurring in Atlantic slope river basins of the eastern U.S., presents a promising opportunity to test several key evolutionary hypotheses in freshwater riverscapes. RBS populations occupy a wide range of ecological niches along most of the river basins that they occur, from clear, turbulent mountain streams to slow moving, blackwater coastal plain rivers. This generalist, and seemingly adaptable, species provides a candidate to study how patterns of genetic diversity respond to varying degrees of connectivity in river systems and the environmental gradients that occur within and among basins. In this study, I used a RADseq approach to genotype and examine variation at thousands of genomic markers across RBS collected from three environmentally distinct groups of ecoregions in each of four replicate river basins. As such, this is the first population genomic study of RBS and one of the first population genomic studies of any sunfish (Centrarchidae). My two broad goals were to (1) test hypotheses regarding the spatial scaling of population structure and the influence of landscape characteristics, particularly barriers, distance, and environmental gradients, in shaping the distribution of genetic diversity and gene flow, and (2) identify genetic markers – through a combination of outlier tests and genotype environment association (GEA) analyses – that were potentially under selection from measurable climatic and physiographic features, and therefore part of the genomic architecture that allows RBS to adapt to such a wide range of environments. I postulate that the findings from this study may provide a baseline for comparison with other similar studies in order to assess for generalities in patterns.

Genetic Diversity: Its Importance and Regulators

Biodiversity is a fundamental component of healthy ecosystems. As such, a decrease in biodiversity can lead to degradation of ecosystems and the services they provide (Mace, Norris et al. 2012). Genetic diversity is the intraspecific component of biodiversity, making its characterization and preservation of primary concern for biologists and conservationists. Biotic and abiotic habitat conditions,

as well as the dispersal of individuals among populations, shape the distribution of genetic diversity through the evolutionary processes of selection, gene flow, and drift. Selection occurs when phenotypes, controlled by underlying genes, show differential fitness under local selection pressures and leads to local adaptation (Chevin, Lande et al. 2010). Selection acts to increase genetic differentiation between populations if selection pressures differ, but only at the genes controlling adaptive phenotypes. Selection also reduces genetic variation within populations. Gene flow, facilitated by dispersal among populations, tends to decrease genetic differentiation between populations and increase genetic diversity in the receiving populations. In contrast, genetic drift causes a decrease in diversity within populations and more genetic differentiation between populations. Drift acts on all populations to some degree but is especially prominent in populations with a low effective population size (N_e). As such, a higher level of genetic diversity is indicative of a population larger size. In sum, low gene flow, high genetic drift, and varied selection pressures in different populations can all result in more genetic differentiation between populations.

Identifying features of the landscape that shape the distribution of genetic diversity is therefore fundamental to our understanding of evolution. Freshwater biota are of particular interest here as dispersal routes are intrinsically bound by the landscape, especially in rivers. Rivers are arranged in “tree-like” networks that feature continuous abiotic and biotic variation along their courses (Hughes, Schmidt et al. 2009). Populations are connected via the river network with varying degrees of gene flow between populations. The simplest model here is that of one-dimensional isolation by distance (IBD), which predicts that the genetic differentiation – often measured using the statistic F_{ST} – is expected to be a linear function of geographic distance between populations (instream distance for river biota) (Wright 1943). However, often further regulators of dispersal between populations can better explain observed genetic structuring, such as obvious dispersal barriers including waterfalls and hydroelectric dams (Roberts, Angermeier et al. 2013, Eschenroeder and Roberts 2020). By testing models incorporating barriers within the context of IBD we can determine how these barriers influence genetic differentiation between populations (i.e., what is their effect on gene flow?)

Evolutionary biologists and conservationists alike are particularly interested in aquatic species' capacities to respond demographically and evolutionarily to global climate change, which is expected to rapidly and fundamentally alter the temperature and hydrology of streams and rivers. Freshwater species are especially at risk due to their limited dispersal possibilities resulting from natural channel boundaries, requiring species to adapt *in situ* (Lynch, Myers et al. 2016). Already-stressful situations arising from sedimentation, pollution, eutrophication, water extraction, hydro-electric dams, habitat degradation, and invasive species combined with the altered temperatures, precipitation, and weather patterns of current and future climate change are expected to fundamentally alter selection regimes faced by species (Palumbi 2001, Ficke, Myrick et al. 2007). In the temperate region of eastern North America, ambient air temperatures have been on the rise since the 1970s and are expected to rise 1-7°C by 2100 depending on greenhouse gas emission rates; this trend is expected to continue even if greenhouse gas emissions are reduced or eliminated (Ficke et al. 2007, Lynch et al. 2016). Along with increased water temperature, changes in precipitation are expected to produce extremes in both drought and flooding periods, resulting in novel hydrologic conditions (Craig 2016). The effects on discharge are difficult to forecast, as the interaction of increased temperatures and increased atmospheric CO_2 may have unpredictable effects on evapotranspiration due to variation in plant responses (Kim, Band et al. 2017). All of these factors and more will play a role in whether populations can persist in the confined landscape of rivers. Greater standing genetic diversity may help populations persist by providing the substrate for novel selections pressures to act on.

Population Genomics

Recent advances in genome sequencing are particularly useful for the study of non-model species, as they allow simultaneous characterization of thousands of loci, the majority of which are presumably neutral (i.e., not under selection) and demonstrate demographic influences of drift and gene flow on genetic diversity and differentiation. A much smaller portion of the loci returned by these approaches are putatively adaptive loci (or loci linked to adaptive loci), which reveal potential evolutionary responses to environmental conditions (Andrews, Good et al. 2016). For example, a series of recent population

genomics studies of fishes found in Australia's large, climatically-variable, Murray-Darling basin provide insight as to the role of spatial and environmental heterogeneity in shaping genetic diversity and divergence (Brauer, Hammer et al. 2016, Harrisson, Amish et al. 2017, Attard, Brauer et al. 2018, Brauer, Unmack et al. 2018). Brauer et al. (2018) found that the distribution of genetic diversity was primarily influenced by dendritic stream structure at neutral markers, showing the influence of dispersal barriers on gene flow, and climatic variation at adaptively-linked markers in the Murray River rainbowfish (*Melanotaenia fluviatilis*). Indeed, after accounting for neutral hierarchical structure, seasonal variation in flow, precipitation, and temperature explained the majority of variation at adaptive markers among streams (Brauer et al. 2018). A similar study in Murray cod (*Maccullochella peelii*) found similar results with the authors suggesting that the translocations of individuals locally adapted to drier, warmer temperatures may increase the probability of future persistence (Harrisson et al. 2017).

Redbreast Sunfish

Redbreast sunfish (*Lepomis auritus*; RBS), a fish native to river basins of the eastern North America, makes an exemplary case study for understanding demographic and evolutionary responses to widely distributed populations and varied local conditions. Their native range extends along the Atlantic slope from southern New Brunswick, Canada to central Florida and in the Gulf slope west to the Choctawhatchee basin. This distribution includes at least 12 EPA level III ecoregions, which differ widely in geology, topography, watershed vegetation types, and human influence (Omernik and Griffith 2014). These generalist invertivores occupy freshwater habitats that span a wide range of environmental conditions, ranging from small, cold, clear, turbulent, high-gradient mountain streams to large, warm, turbid, placid, low-gradient coastal plain rivers, and are common throughout these habitats (Aho, Anderson et al. 1986, Jenkins and Burkhead 1994, Benke and Cushing 2011). The basins occupied by RBS typically begin as mountain streams over high-relief, metamorphic and igneous geology in the Appalachian province and terminate in the sedimentary low-relief coastal plain. The primary substratum of the mountainous regions include gravel, cobble, and bedrock, whereas the coastal plain regions are largely comprised of sand, with some silt and detritus. Due to variation in the underlying geology, streams

of similar size in these regions show contrasting morphology. For example, in mountainous regions streams are typified by relatively straight channels with a longitudinal pattern of riffles, runs, and pools comprising most habitat, whereas coastal plain streams are generally meandering with log jams and wide floodplains providing the majority of habitat (Benke and Cushing 2005). Hence, RBS are likely more limited in optimal habitat and dispersal in mountainous regions as they prefer pools that are often small and bordered by inhospitable, shallow riffles; mountainous regions are also typified by complete dispersal barriers in the form of waterfalls and rapids. In addition, suitable spawning habitat (0.1-0.5 cm particle size) is also less common in mountainous regions (Aho and Anderson 1986). In contrast, the coastal plain is largely hospitable to RBS with fewer dams and an abundance of sand and small gravel for nest construction. The Piedmont province is located between these two geographic extremes, and presumably is intermediate in characteristics though more similar to the underlying geology (i.e. metamorphic rock and intrusive granite) of the mountains. Rivers in this region generally have the highest turbidity (suspended solids) due to higher relief compared to the coastal plain and more erosive soils than the mountainous regions (Simmons 1976). The piedmont is also the location of most hydro-electric dams on these basins, which have led to altered hydrologic regimes, cooler temperatures in tailrace areas, sediment trapping in reservoirs, and represent strong dispersal barriers for most macroorganisms (Benke and Cushing 2005).

Besides variation in other physicochemical characteristics, variation in the latitudes and elevations occupied by redbreast sunfish populations lead to variation in local temperature. Temperature plays a role in metabolism adaptation, especially in ectothermic species such as redbreast sunfish. Specifically, fish at lower temperatures tend to have higher metabolic rates, likely the result of shorter growing seasons requiring more rapid growth or increased growth at lower temperatures (Yamahira and Conover 2002). Also, photoperiod variation due to latitude may also play a role in phenology and circadian genes have been found to vary along North-South clines in *Oncorhynchus tshawytscha* (O'Malley and Banks 2008).

Hypotheses

In the present study I have assessed population genomic variation within and among RBS populations in mountain, Piedmont, and coastal plain regions of four major river basins. The physical/chemical differences among these locations, as well as the dendritic network connecting them presumably have exerted strong influences on the demography and evolution of redbreast sunfish populations occupying these ecoregions. Demographically, I hypothesized that mountain populations would be numerically smaller than coastal plain populations because mountainous regions would exhibit less available habitat and smaller habitat patches because of more-frequent dispersal barriers (e.g., fast and turbulent flows, shallow riffles, waterfalls, dams), and that as a result, genetic diversity would be lower and gene flow would be more restricted (i.e., steeper IBD) in these areas than in coastal plain habitats (Maia et al. 2015). I also expected RBS populations in general to show hierarchical population structure due to strong divergence among basins, and among ecoregions within basins, with more moderate IBD over smaller scales within connected, environmentally homogeneous habitat patches (Hughes, Schmidt et al. 2009). Moreover, I expected to find a trend of increased within-basin, neutral genetic diversity in the downstream direction as the result of downstream-biased dispersal causing a pattern of higher genetic diversity at sites located nearer to the river outlet. Downstream-biased dispersal may be expected as it is energetically efficient to disperse downstream and larva are likely to be moved downstream (Paz-Vinas, Loot et al. 2015). Alternatively, genetic diversity might be equally high upstream if upstream dispersal counteracts this phenomenon.

From an evolutionary standpoint, I hypothesized that varied environmental selection pressures in mountain versus coastal-plain habitats have produced adaptive genetic variation. For example, when examining adaptive loci, I predicted that across replicate river basins, populations in more climatically similar ecoregions (e.g., mountain sites with similar temperature and precipitation regimes) would exhibit consistent genomic signatures of adaptation to these climatic regimes, as indicated by (1) ecoregional convergence on similar alleles, regardless of basin, and (2) a stronger signal of genetic isolation-by-environment (IBE: ecoregions explain more variation in FST) than isolation-by-distance (IBD: basins

explain more variation in FST). I used multivariate genotype-environment association (GEA) analyses to explicitly link alleles to the environmental variables hypothesized to represent variable selection pressures in redbreast sunfish (Forester et al. 2018). Environmental associations of candidate adaptive loci were corroborated by mapping these loci to annotated genomes, to see whether they linked closely to functional genes of relevance to local adaptation.

CHAPTER 2: METHODS

Site Selection and Sample Collection

Between 8 July 2017 and 27 September 2018, I sampled a total of 32 sites and collected 146 individual RBS for genetic analyses. RBS were sampled via a factorial design spanning three ecoregion groups (mountain, Piedmont, coastal plain) within each of four major river basins [Apalachicola-Chattahoochee-Flint (ACF), Savannah (SAV), Roanoke (ROAN), James (JAMES)] (Figure 1). The sampling design included three mountain and coastal plain sites and two Piedmont sites per basin. The Southeastern Plain, Southern Coastal Plain, and Middle Atlantic Coastal Plain ecoregions, as delineated by Omernik and Griffith, 2014, were classified as "coastal plain." Similarly, the North Central Appalachians, Blue Ridge, and Ridge and Valley ecoregions were classified as "mountain" (Table 1; Figure 1).

Sites were selected by locating access points near public rights-of-way (e.g., traffic bridges) via Google Earth with an eye towards selecting small to mid-sized streams distributed evenly within each ecoregion (i.e., not clumped). Elevation was derived from *StreamStats* version 4 (U.S. Geological Survey 2016) and was used to support the ecoregion membership of sites, within the following elevation categories: coastal plain= 0-50 m, Piedmont =50-250 m, mountain >250 m. Initially, I sought to sample sites of consistent stream-size (i.e., with similar upstream drainage areas). However, this proved difficult due to consistent differences in the stream size-drainage area relationship among ecoregions (e.g., small mountain streams tended to have less drainage area than comparable size coastal plain streams). Also, the native range of RBS includes basins further west and north of the basins sampled. However, the sampled basins represent the western-most (ACF) and northern-most (JAMES) basins in which all three ecoregions are clearly defined and RBS are common.

Fish were captured via backpack electrofishing or angling, depending on site depth and water transparency. At least three and up to seven individual fish were collected at each site, per Georgia Southern University Institutional Animal Care and Use Committee Protocol #I17010. Fish were euthanized in a 100 mg/L solution of buffered MS-222, and then a pectoral fin clip was collected and placed in 95% ethanol at -20°C. The remainder of each fish was preserved in a 10% formalin solution and later transferred to 70% ethanol for a concurrent morphological study. Sample sizes varied due to difficulties in sampling and/or low densities of RBS.

Environmental and Geographic Data

Positional data was collected at each site via GPS. Using the U.S. Geological Survey's *The National Map* web application, the shortest in-stream distance between each pair of sites was derived manually via tracing with the distance measuring tool along the National Hydrology Dataset layer. The shortest instream distance between each site and its respective basin outlet was also measured. Any major dams or reservoirs between sites were also recorded during this process in order to form an ordinal scale of between-site connections. Specifically, site-pair connections were recorded as "ocean", "dam", "impoundment", or "riverine"; the highest resistance barrier (i.e., ocean) was recorded when site-pairs had multiple connection types. In addition, a site-pair connection was considered "impoundment" when the sampled streams flowed into a common impoundment but were not separated by the downstream dam that produced the impoundment. I also traced along the coast line in order to determine the shortest waterway distance between basin outlets. Also within *The National Map*, I derived a local gradient (GRADIENT; meters per kilometer) using the spot elevation tool and distance measuring tool. Specifically, I traced 1 km upstream and downstream of the site (2 km total) and recorded elevation at endpoints. If a dam was present in 2 km stretch the downstream side of dam was the start point of the 2 km stretch. I calculated straight-line distance (i.e. "crows-fly") between sites in R (R Developement Core Team 2021), using the package *geodist* (Padgham and Sumner 2021) and the "geodesic" option.

The U.S. Geological Survey's *StreamStats* version 4 web application was used to extract environmental characteristics about the watershed upstream of each site. This application delineates the

watershed boundaries upstream of a user-defined point and returns characteristics within the delineated basin such as land cover, drainage area, and elevation. Specifically, I used *StreamStats* to derive the minimum basin (i.e., site) elevation (ELEVMIN; meters) and the area that drains to a site (DRNAREA; square kilometers). DRNAREA was used as a proxy for stream-size in this study. Data consisting of the percentage of land-use from NLCD 2006 classes 21-24 (LCDEVEL; human developed areas with various degrees of impervious land cover) and the percentage of forests and shrub lands from NLCD 2006 classes 41-52 (LCFORSHB; Deciduous Forest, Evergreen Forest, Mixed Forest, and Shrubland) was also extracted (U.S. Geological Survey 2011). As a caveat, the LCFORSHB data was not available for all sites so data for NLCD 2006 classes 41-43 (Deciduous Forest, Evergreen Forest, Mixed Forest) was used instead under the assumption that shrublands do not make up a significant proportion of the watersheds studied.

Climatic and soil characteristics in the vicinity of each site were extracted from the *EarthEnv* freshwater-specific environmental variables (Domisch, Amatulli et al. 2015). This global dataset compiles data at the grain of 1-km pixels, summing conditions across all pixels in the watershed upstream of each site. I extracted the following for use as environmental predictor variables: inverse distance-weighted annual mean upstream temperature (MEANTEMP; °C * 10), upstream temperature seasonality (TEMPSD; standard deviation *100), inverse distance-weighted sum of annual upstream precipitation (PRECIP; millimeters), upstream precipitation seasonality (PRECIPCV; coefficient of variation), and soil cation exchange capacity (CATIONEXCH; cmol per kilogram). I considered the latter a potential surrogate for limestone geology and thus groundwater influence. The source data for MEANTEMP, TEMPSD, PRECIP, and PRECIPCV were derived from *WorldClim* datasets (Hijmans, Cameron et al. 2005) and data for CATIONEXCH was sourced from an ISRIC dataset (Hengl, de Jesus et al. 2014). Initially, a larger set of many additional variables were extracted from *EarthEnv*. However, after observing collinearity among groups of variables, the listed variables were selected for further downstream analyses due to their interpretability.

For each environmental variable, I calculated the difference in the observed value between each pair of sample sites, then standardized each resulting distance measure to have a mean of zero and standard deviation of one. These pairwise environmental-distance measures were used along with spatial-distance measures in comparisons with genetic distance measures (see below). In addition, two composite environmental-distance variables were made: 1) Physiographic distance = the sum of pairwise differences using ELEVMIN, GRADIENT, CATIONEXCH, LCDEVEL, and LCFORSHB, and 2) Climate distance = the sum of pairwise differences using MEANTEMP, TEMPSD, PRECIP, and PRECIPCV.

In order to visualize how sites separated in multivariate environmental space, a PCA of all site point measures was performed using the R package *ade4* (Dray and Dufour 2007). I assessed whether ecoregions, basins, or ecobasins formed clusters and plotted the individual variable loadings in order to visualize the contribution of individual variables to the PCs.

DNA Extraction

Whole genomic DNA was extracted from fin clips via the DNeasy Blood & Tissue kit (Qiagen, Germany) with an overnight lysis period. Approximately 10 mg of fin clip material was used in this spin-column method after allowing the ethanol to evaporate. DNA quality was checked on 1% agarose gels, ensuring a band of high molecular weight for each extract. DNA concentration was quantified via Qubit Broad Range assays (Life Technologies, USA) and normalized to 20 ng/µl prior to library preparation.

Genomic Library Preparation & Sequencing

A single genomic library including all sampled individuals was created following the 2RAD protocol (Hoffberg, Kieran et al. 2016, Bayona-Vásquez, Glenn et al. 2019). This protocol is a modification of the original double digest Restriction site Associated DNA sequencing (ddRADseq) method which is designed to create reliable, reduced-representation genomic libraries at low cost without requiring previous genomic knowledge (Peterson, Weber et al. 2012). ddRADseq methods use the sequence specificity and DNA cleaving properties of two restriction enzymes to generate homologous DNA fragments among individuals and size selection to further reduce the pool of fragments for sequencing. These methods also incorporate sequence “barcodes” so that many individuals can be

multiplexed and sequenced simultaneously. The 2RAD method was specifically chosen due to the ability to remove PCR duplicates as they can lead to inflation of homozygosity when there is a bias in the PCR amplification of alleles (i.e., they can lead to allele drop out) (Schweyen, Rozenberg et al. 2014). 2RAD does so via the incorporation of an eight-nucleotide degenerative base region onto each fragment prior to PCR amplification which allows for removal of duplicates downstream (see below). Additionally, 2RAD uses molecular adapters and barcoding primers available at relatively low cost (baddna.uga.edu) which are designed for reliable use with standard Illumina sequencing protocols.

The 2RAD protocol was followed as per Bayona-Vásquez, Glenn et al. (2019) with the following modifications: 1) Restriction enzyme digests were performed for two hours instead of one. 2) AMPure XP beads (Beckman Coulter, Brea, CA, USA) were used instead of SpeedBeads for purification. 3) Eight PCR cycles were used instead of six. After restriction enzyme digest, barcoding, PCR amplification, and several purification steps, fragments were size selected for 525 base-pairs using a Pippin Prep (Sage Science, Beverly, MA, USA) and the "tight" setting. The resulting library was sent to the Genomics and Cell Characterization Core Facility at the University of Oregon and sequenced on a single Illumina HiSeq 4000 lane, using 150-bp paired-end reads and dual 8-bp index reads.

Locus Assembly & SNP Calling

The *de novo* assembly of the 150-bp reads into loci was performed using the STACKS pipeline version 2.3e (Rochette, Rivera-Colón et al. 2019). First, the *process_radtags* module was used to demultiplex the raw Illumina data. During demultiplexing, reads were also trimmed to 140 base pairs and any read with an uncalled base was removed. Additionally, reads with low Illumina quality scores were removed using the default sliding window method as implemented in *process_radtags*. Following demultiplexing, the *clone_filter* module was used to remove PCR duplicates using the degenerative base region incorporated during library preparation. This module detects PCR duplicates when the degenerative base region and the primary sequence match for two or more reads and retains only one of the duplicate reads per individual.

Following demultiplexing, cleaning low quality reads, and PCR duplicate removal, the reads were ready for loci assembly in the primary STACKS pipeline. The *de novo* assembly of loci was used here as there is no reliable reference genome for RBS. This process started with *ustacks* forming "stacks" of identical reads within individuals. I set a minimum stack size of three reads (m = 3) and stacks with lower coverage than this were excluded. Next, the stacks were merged with similar stacks according to the number of base pair mismatches allowed by the M parameter to form individual-level loci. I set this to three and kept all other *ustacks* parameters on default settings. Next, the *cstacks* module was used to merge individual-specific loci into homologous loci across the individuals if they differed by less than three base pairs (n = 3). This formed a catalog of loci encompassing all individuals and all loci determined in *ustacks*. Of note, I tested a range of loci assembly parameters in STACKS as per Paris et al., 2017, including m = 2-10, M = 1-8, N = 3-10, and n = 1-8. The final parameter set (m = 3, M = 3, N = 5, and n = 3) was chosen due to its similarity to other studies and a lack of benefit to parameters at either extreme of the tested range with regards to the number of polymorphic loci produced.

With all loci cataloged, the STACKS *populations* module was implemented to produce a set of SNPs that were consistently present in a majority of individuals across a majority of sites. *Populations* allows the user to define *a priori* populations and filter loci based on their presence in a minimum number of populations and/or a minimum percentage of individuals at each population. I first defined populations as individuals from the same basin and ecoregion (e.g., three Roanoke River mountainous sites collapsed into "Roan-MT") resulting in twelve "eco-basin" populations. I then required that a locus be present in at least 70% of individuals (r = 0.7) within at least 10 of the 12 ecobasins (p = 10). I also implemented a minimum minor allele frequency of 0.05 and retained only the first SNP per locus to avoid linkage. The program PGDSpider 2.1.0.0 (Lischer and Excoffier 2011) was used to convert the resulting STACKS dataset into formats required for downstream analysis programs.

Outlier SNP Detection

I used a series of "outlier detection" tests to separate my larger SNP dataset into two smaller datasets: 1) putatively neutral SNPs that were useful for testing demographic questions about migration

and drift and 2) “outlier” SNPs that were potentially under selection and thus potentially useful for answering evolutionary questions about local adaptation. I used four outlier detection methods to identify SNPs that did not meet the assumptions of selective neutrality. Each outlier detection method was performed on the full SNP dataset so that overlap between detection methods could be determined. First, locus by locus Hardy-Weinberg equilibrium (HWE) exact tests were conducted in Arlequin 3.5.2.2 (Excoffier and Lischer 2010), grouping individuals into 12 ecobasins and employing 10^6 Markov-Chain-Monte-Carlo (MCMC) iterations, following a burn-in of 10^5 iterations. A SNP was considered a HWE outlier if it failed the test ($p < 0.05$) in greater than half of the total tests conducted (maximum of 12 tests possible). Next, *Bayescan* (Foll and Gaggiotti 2008), a Bayesian *Fst*-based outlier test, was used to identify outlier loci which showed much greater or much lower between population differentiation than the average locus. In *Bayescan* I set the prior odds to 10000 (default) and the false discover rate (FDR) threshold to 0.1 and performed 100000 simulations. I used another differentiation-based outlier detection method in the R package *pcadapt* (Luu, Bazin et al. 2017). *pcadapt* first defines population structure based on principal component analysis (PCA) then identifies outliers based on atypical correlations to the inferred population structure. I kept the first two PCs as they occurred before the asymptote in the scree plot of eigenvalues. I used the Benjamini-Hochberg procedure with a FDR of 0.1 to call outliers in *pcadapt*.

The three previous methods of outlier detection only consider allele frequencies across *a priori* or PCA-inferred populations and do not consider the covariation between environmental variables and allele distribution. In order to explicitly incorporate the possible associations between multiple environmental characteristics and multilocus allele frequencies, I used a constrained ordination in the form of redundancy analysis (RDA) (Forester, Lasky et al. 2018) within the R package *vegan* (Oksanen et al. 2015). For environmental predictor variables I first performed PCA on MEANTEMP, TEMPSD, PRECIP, PRECIPCV, GRADIENT ,ELEVMIN, LCDEVEL, LCFORSHB, CATIONEXCH and retained the first two PCs (explained 65% of the variation) as predictors in the RDA. I also used DRNAREA as a predictor to see if this proxy for stream size could explain environmental associated variation.

Importantly, I performed two separate RDAs, one for the two southern basins (ACF,SAV) and one for the two northern basins (ROAN,JAMES), after noting the degree of differentiation between these two groupings in preliminary PCAs of the whole genomic dataset. A locus was considered an outlier if its correlation with any of the first three RDA (environmental) axes was more than three standard deviations from the mean; for these loci, I then recorded which environmental predictor exhibited the strongest correlation with that locus.

After the four outlier-detection methods were completed, I constructed two distinct genomic datasets: one putatively neutral and one with putatively adaptive outlier SNPs. For outlier loci I used a conservative approach by including only loci that failed HWE or were detected in either the north or south RDAs, but not including loci that were only detected in either *Bayescan* or *pcadapt*. The neutral dataset, in contrast, comprised the remaining loci, including those detected by *Bayescan* or *pcadapt* but no other method.

Delineating Population Structure

I used a series of complementary methods to assess the juxtaposition and spatial scaling of population structure, using the neutral SNP dataset. Analysis of Molecular Variance (AMOVA) was performed to estimate how much molecular variance emerged at a series of discrete spatial scales: among basins, between ecoregions within basins, between sites within ecobasins, and within sites. The AMOVA was conducted using the R package *poppr* (Kamvar, Tabima et al.) and the *randtest.amova* function to verify the covariance components (10000 permutations). This was complemented by a principal components analysis, which did not assume discrete population structure. PCAs were conducted in the R package *adgenet* (Jombart and Ahmed 2011) with missing data being encoded as the mean genotype for a given sample site. PCA was used to visualize the data and observe whether distinct clusters could be discerned and whether the clusters corresponded to ecobasin-, basin-, or ecoregion-level structuring. Finally, I used a model-based Bayesian clustering algorithm, implemented in STRUCTURE v2.3 (Pritchard, Stephens et al. 2000), to infer the true number of genetic clusters (i.e., populations) in the dataset and assign individuals to those clusters. STRUCTURE models allowed for admixture and

correlated allele frequencies and did not utilize spatial information as a prior. I assessed the likelihood of anywhere from 1 to 10 clusters (i.e., K=1-10) with 5 replicate models runs for each K value; each run comprised a burn-in period of 10^5 iterations, followed by 10^6 Markov chain Monte Carlo steps. Following Evanno, Regnaut et al. (2005), I used mean log-likelihood and judgement (see Results) to define the most-likely K and created individual ancestry bar plots with the online tool CLUMPAK (Kopelman, Mayzel et al. 2015). I first performed STRUCTURE on the whole neutral dataset, and then based on these results, within the northern (JAMES and ROAN basins) and southern regions (ACF and SAV) separately, in order to delineate hierarchical structure in the genomic data.

Landscape Influences on Genetic Diversity and Gene Flow

I calculated observed (H_O) and expected heterozygosity (H_E) for each site using the *hierfstat* (GOUDET 2005) package in R. Expected heterozygosity (H_E) is a common and unbiased measure of genetic diversity and was used as the response variable in models testing hypotheses regarding environmental influences on genetic diversity. I performed a linear mixed effects model in the R package *lme4* (Bates et al., 2015) to determine if genetic diversity was significantly different among ecoregions while controlling for the effect of basin using a random effect.

Pairwise population differentiation was assessed with the fixation index (F_{ST}) (Weir and Cockerham, 1984). This measure of genetic distance was calculated between all sampling sites in the program *ARLEQUIN* v.3.5 (Excoffier and Lischer 2010), and significance determined by 10,000 permutations. F_{ST} is inversely related to gene flow, and was therefore used as the response variable for models testing the influence of geographic and environmental features on genetic differentiation and gene flow. I used multiple regression on distance matrices (MRM) in order to test whether each of the two pairwise response variables, neutral FST and outlier FST, were more strongly associated with spatial distance (IBD) or with environmental distance (IBE) separating sites. I hypothesized that neutral FST would be more strongly associated with spatial than environmental distance, and the opposite to be true for outlier FST. I indexed spatial distance as the along-the-stream distance between sites and environmental distance using the climate and physiographic distance measures described previously.

These relationships were tested using MRMs in the R package *ecodist* (Goslee & Urban, 2007), with 10,000 permutations to test significance. Importantly, MRMs can compare greater than two distance matrices, so I included both environmental distances (physiographic and climate) and in-stream distance as predictors in each model and calculated the partial R^2 for each. This allowed me to test whether climate or physiography explained a significant additional component of genetic variation, after accounting for the effect of spatial distance and IBD. Spatial, climate, and physiographic distance measures were standardized for this analysis, so that their estimated effect sizes were on the same measurement scale (mean = 0, standard deviation = 1) and could be directly ranked. I also performed MRMs on within basin comparisons to assess IBD on a per basin basis. Some MRMs were plotted and visually assessed for trends that could be associated with population dynamics.

One of my specific predictions was that, all else being equal, gene flow would be greater among coastal plain than mountain sites, because of the higher gradient and resistance of montane landscapes. To test this prediction, in a subset of pairwise comparisons for which there were no reservoirs, dams, or basin boundaries between sites (i.e., controlling for the effects of barriers) I tested the effect of in-stream distance on genetic distance via a linear model in the R function *glm*. Specifically, I tested whether the slope and/or intercept of IBD varied significantly between ecoregions. Because this was a relatively small dataset, I did not include a random effect for basin.

A random forest (RF) modeling approach was used to measure the relative importance of site and between-site environmental attributes in predicting genetic diversity within populations and genetic differentiation/gene flow among populations. Because it is a non-parametric method that easily accommodates nonlinearity, collinearity, and variable interactions, the RF modeling approach is a promising method to detect the influence of the landscape in shaping the genome of a widely distributed species such as RBS. RF modeling uses a machine learning algorithm to create decision trees which are “grown” by selecting a random subset of predictor variables and testing which predictor variable can split the data into the two most homogeneous groups. This splitting process is repeated until all variation has been accounted for or all levels of the predictor variables have been used. Rather than focusing on a

single best decision tree, RF builds thousands of alternative trees based on bootstrap subsets of predictor variables, thus measuring variable importance by how well the predictive power of decision trees is affected by the presence or absence of that variable.

RF models were built using the *randomForest* package in *R* (Liaw & Wiener, 2002; R Core Team, 2019). Two separate sets of models were tested: 1) Response as pairwise genetic divergence (F_{ST}) and predictors as pairwise site differences [BARRIER (highest resistance barrier between sites), BASIN (for the within basin models only=basin that the comparison occurred in), instream.distance.km (distance between sites along the stream) and 2) Response as genetic diversity (H_E) and predictors as point measures [distance.ocean (distance along the stream to the river outlet), BASIN (basin the site was located in), ECOREGION (ecoregion classification of a site), ECOBASIN (combination of BASIN and ECOREGION), and all variables listed in Table 2]. For genetic divergence models I predicted that BARRIER would be positively related to F_{ST} with increasing differentiation occurring in the order of : riverine-impundment-dam-ocean, and I predicted instream.distance.km to reflect a positive IBD relationship. For within basin models, I predicted BASIN to have little effect on within-basin F_{ST} indicating roughly equal population differentiation occurring within basins. In genetic diversity models, I predicted distance.ocean to show a negative relationship with H_E indicating a downstream increase in genetic diversity. I also predicted BASIN and ECOBASIN to have little effect, while ECOREGION was predicted to show a positive relationship with H_E with increased diversity in the coastal plain, potentially due to larger populations in this high quality habitat. For environmental point measures, I predicted a positive relationship with diversity for MEANTEMP perhaps due to higher growth and thus higher N_E at higher temperatures. LCDEVEL and LCFORSHB were predicted to have opposite effects, were I expected a positive relationship between diversity and LCFORSHB but a negative relationship with LCDEVEL in accordance with the effect of anthropogenic impact reducing population sizes. For all models, I set the number of trees grown (*ntree*) to 5000 and the number of variables to test at each split (*mtry*) was set to 1 for pairwise models and 2 for point measure models. I then calculated the variance importance score for each predictor using percent increase of mean squared error (%incMSE). This is

calculated by measuring the percent increase in MSE when the predictor variable in question is effectively removed from the model by randomly permuting its values among observations. I considered predictors with the highest %incMSE to be most important for predicting the response variable (F_{ST} or H_E). Furthermore, to find the best fitting model I removed predictors that showed collinearity and any that had an importance score <10%. Overall model performance was assessed using coefficient of determination (R^2). Models with the highest R^2 were selected as the best performing model and partial dependence plots were created using *pdp* (Greenwell, 2017) in R. Models for HE and FST were both fit using all observations (i.e., all sites for HE and all pairs of sites for FST). Because basin membership was confounded with the presence/absence of an ocean barrier, and given that I wished to assess the influence of basin on FST and gene flow, I also built an RF model for FST consisting only of pairwise comparisons within the same basin (i.e., excluding any comparisons in which barrier type = "ocean").

Functional Annotation of Adaptive Loci

In a largely exploratory analysis, all SNP loci were queried for possible gene annotation and corresponding gene ontology (GO) terms. Consensus DNA sequences were exported from STACKS and a BLAST search was performed in BLAST2GO (Conesa et al., 2005) using the nonredundant nucleotide database (BLASTX). I only considered BLAST hits that were above an e-value threshold of 1×10^{-3} and set an annotation e-value threshold of 1×10^{-6}. In order to assess whether the outlier SNPs were more likely than neutral SNPs to BLAST to annotated functional genes, enrichment of GO terms in the outlier data set was assessed relative to all SNPs using Fisher's exact test with a FDR of 0.05 in BLAST2GO. I used CATEGORIZER (Hu, Bao, & Reecy, 2008) to summarize GO terms assigned to the outlier loci according to the GOSlim classification method. This simplification of possible gene function was done in order to see if any major trends were present in the outlier loci that may show biological relevance.

Table 1. Names, locations, and codes of sites sampled in this study. Numbers in parentheses represent individuals remaining after removal of outlier individuals.

Site Code	Basin	Ecoregion	Site Name	Latitude	Longitude	Individuals
A1	ACF	CP	Aycocks Creek	31.1085	-84.7644	4
A2			Sawhatchee Creek	31.1808	-85.0435	4
A3			Cooleewahee Creek	31.4052	-84.3001	4
A4		PD	Centralhatchee Creek	33.3111	-85.1047	5
A5			New River	33.2489	-84.9353	4
A6		MT	Soque River	34.6182	-83.5294	5(4)
A7			Chattahoochee River	34.6873	-83.7101	4
A8			Chestatee Creek	34.5278	-83.9400	5
S1	Savannah	CP	Beaverdam Creek (A)	32.8213	-81.6229	4
S2			Beaverdam Creak (B)	32.9373	-81.8152	4
S3			Ebenezer Creek	32.3648	-81.2315	3
S4		PD	Horn Creek	33.6642	-82.0679	4
S5			Steven's Creek	33.7946	-82.1454	4
S6		MT	Chauga River	34.8331	-83.1748	4(3)
S7			Chattooga River (A)	34.9734	-83.1158	5(1)
S8			Chattooga River (B)	34.8154	-83.3066	5
R1	Roanoke	CP	Lee's Creek	36.4931	-77.6297	7
R2			Roanoke River side channel	36.4321	-77.5872	6
R3			Quankey Creek	36.3183	-77.5947	7
R4		PD	Goose Creek	37.2666	-79.5875	5
R5			Roanoke Creek	36.9309	-78.6650	5
R6		MT	North Fork Roanoke River	37.1869	-80.3525	5
R7			South Fork Roanoke River	37.1625	-80.2418	3
R8			Tinker Creek	37.2846	-79.9196	3
J1	James	CP	Chickahominy River	37.5757	-77.3367	4
J2			Falling Creek	37.4393	-77.4391	5
J3			Swift Creek	37.3139	-77.4943	5
J4		PD	Hardware River	37.8125	-78.455	5
J5			Willis River	37.6670	-78.1655	5
J6		MT	John's Creek	37.5064	-80.1060	5
J7			Dunlap Creek	37.8008	-80.0473	4
J8			Calfpasture River	37.9670	-79.4966	4

Table 2. Environmental predictor variable codes, sources, explanations, and units. The "Category" column indicates the environmental distance measure in which that variable was included.

Variable Code	Source Name	Source	Unit	Details	Category
ELEVMIN	MINBELEV	Streamstats	meters	Minimum basin elevation	Physiographic
DRNAREA	DRNAREA	Streamstats	square kilometers	Area that drains to a point on a stream	
GRADIENT	NA; manually derived	USGS National Map	meters per kilometer	Traced 1km u.s. & d.s. of site (2 km total) and recorded elevation at endpoints	Physiographic
MEANTEMP	hydro_wavg_01	EarthEnv	Celsius * 10	Annual Mean Upstream Temperature; weighted average	Climate
TEMPSD	hydro_wavg_04	EarthEnv	Celsius/ Standard Deviation * 100	Upstream Temperature Seasonality (standard deviation *100)	Climate
PRECIP	hydro_wavg_12	EarthEnv	millimeters	Annual Upstream Precipitation; weighted sum	Climate
PRECIPCV	hydro_wavg_15	EarthEnv	millimeters * 100	Upstream Precipitation Seasonality (Coefficient of Variation); weighted sum	Climate
CATIONEXCH	soil_wavg_07	EarthEnv	cmol per kilogram	Cation exchange capacity	Physiographic
LCDEVEL	LC06DEV	Streamstats	%	Percentage of land-use from NLCD 2006 classes 21-24 (Human developed areas)	Physiographic
LCFORSHB	LC06FORSHB	Streamstats	%	Percentage of forest from NLCD 2006 classes 41-43 (Deciduous Forest, Evergreen Forest, & Mixed Forest)	Physiographic

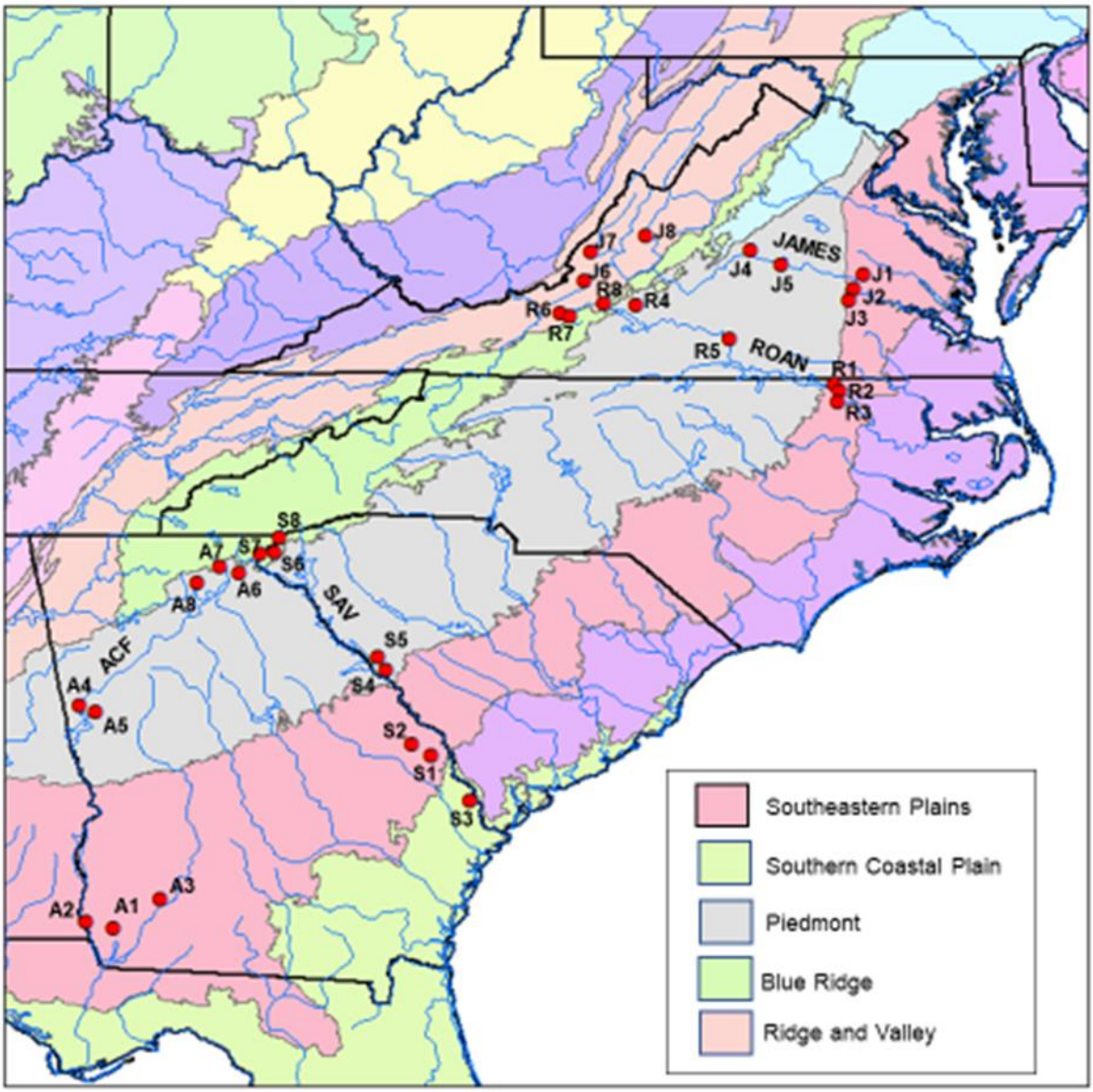

Figure 1. Map of sampling locations within four river basins (ACF = Apalachicola-Chattahoochee-Flint, SAV = Savannah, ROAN = Roanoke, JAMES = James) and five EPA level-III ecoregions. These ecoregions were collapsed into three main ecoregion groups (coastal plain, Piedmont, and mountains) for this study.

CHAPTER 3: RESULTS

Environmental Variation Among Sites

A PCA of all sites using 10 environmental variables (Table 2) resulted in PC axis 1 and PC axis 2 explaining 35% and 26% of the variation, respectively (Figure 2). The loading vectors of the environmental variables combined with grouping of sites into hierarchical categories (ecobasin, ecoregion, and basin) allowed for the characterization of environmental variation at several levels. First, at the finest scale, each ecobasin occupied its own region in the environmental PC space. When grouped by ecoregion, coastal plain, Piedmont, and mountain sites primarily differed along a physiographic axis including higher site elevation, stream gradient, cation exchange capacity, and forested land cover in mountain sites than coastal plain sites, whereas Piedmont sites were intermediate. In contrast, basins primarily differed along a north-south axis of climate, with higher temperatures, higher precipitation, and more variable precipitation in southern basins, but more variable temperatures in the northern basins. Stream size and developed land cover also were higher in the northern basins than southern basins. These trends indicate that, contrary to intentions of my study design, ecoregions were not perfectly replicated across basins, especially with regards to climate variables (precipitation, temperature). Furthermore, a strong correlation (r= 0.60) was found between instream distance and the composite environmental distance variable. The physiographic distance variable was not strongly corelated (r= -0.03) with instream distance. These results suggest that climate differed more by latitude and basin than by ecoregion, whereas physiographic characteristics varied more consistently among ecoregions across basins.

Genome-wide SNP Discovery

This study successfully genotyped RBS sampled from four river basins using the ddRADseq method and the STACKS *de novo* bioinformatic pipeline. After demultiplexing and quality filtering the 525,289,254 reads from the raw Illumina output, I retained a mean of 3,416,338 reads per individual (range: 1,118,197-7,809,875). Following catalog formation in STACKS and after retaining one SNP per locus, I retained 2,858 SNPs for downstream analyses. An initial PCA of these data revealed six individuals that clustered well outside of other individuals from the same site. These included one

individual each from sites A6 and S6 and four individuals from site S7. I can offer no confident interpretation of these outliers, which may have been due to a range of factors from poorer-quality DNA extracts to lower sequencing coverage to genetic introgression with other species (*Lepomis* species frequently hybridize). Nonetheless, to avoid these individuals unduly influencing analyses of interest for this thesis, I removed these individuals from the dataset and excluded site S7 altogether, given that the remaining sample size of one individual from this site was insufficient for representing population genetic characteristics.

Outlier SNP detection

Outliers were detected via all four of the methods I used, but the numbers of outliers varied greatly by method (Figure 3). HWE tests found 36 of the SNP loci did not meet its assumptions in at least half the tests conducted. *Bayescan* showed the highest count of outliers detected (637), though I only considered loci detected via this method outliers when they were also detected in at least one other method (123 loci met this criteria). *Pcadapt* found 110 SNP loci, of which 82 overlapped with other methods and were called as outliers. The highest degree of overlap was between *Bayescan* and *Pcadapt* with a total of 79 SNP loci being called between them.

The northern-basin and southern-basin RDAs discovered 158 and 22 outliers, respectively. Of these, only two loci were detected as outlier in both methods. The PCA of environmental variables showed that the first two components explained a cumulative 65.3% of variation and these components were used in the RDAs along with drainage area (Table 3). Based on loadings from the initial PCA, Comp.1 primarily represented variation in physiographic and land use attributes (ELEVMIN, CATIONEXCH, LCDEVEL), whereas Comp.2 was primarily a climatic PC (MEANTEMP, PRECIP, TEMPSD). I decided to use DRNAREA as a standalone variable in the RDA in order to determine if any SNPs showed evidence of selection corresponding to this proxy of habitat size. In the north RDA I detected 158 outliers with 65, 39, and 62 of these having the strongest correlations with Comp1., Comp.2, and DRNAREA, respectively. The south RDA detected less outliers with its 21 outliers having the strongest correlations with Comp.2 (15 loci) and DRNAREA (7 loci), and no correlations with Comp.1.

Also, 33 RDA outliers overlapped with at least one other method and two loci were detected in both regional RDAs. In total 287 loci were filtered out of the whole genomic dataset to form the outlier dataset, and the remaining 2,571 loci were retained in the neutral dataset.

Genetic Diversity and Differentiation

Overall, H_E ranged from 0.34 to 0.48 across all sites for the neutral dataset (JAMES: 0.38-0.42, ROAN: 0.34-0.48, SAV: 0.37-0.48, ACF: 0.38-0.43; Table 3, Figure 4). A GLMM test for the effect of ecoregion on H_E (using basin as a random effect) was not significant (p=0.191). Although H_E increased from coastal plain to mountain sites in ROAN, this was not the case in other basins. These results do not support the hypothesis of a downstream increase in genetic diversity, as no downstream increase in genetic diversity was found in any of the studied basin.

Pairwise F_{ST} ranged from -0.086 to 0.79 across all comparisons; most low F_{ST} estimates were non-significant (Table 5). In general, F_{ST} was highest in between basin comparisons, with some exceptions such as the comparison between J1 and R3 (-0.007, not significant). In addition, the range of within-basin F_{ST} comparisons varied by basin. The two southern basins had the greatest range with maximums of 0.43 and 0.40 in the Savannah and ACF, respectively. The Roanoke had a maximum of 0.054 and the James 0.11. All basins had F_{ST} comparisons that were essentially zero and nonsignificant, and these primarily occurred within ecobasins. Between basins, the comparisons between the R3 and all coastal plain JAMES sites (J1,J2,J3) were nonsignificant. Between ecobasins, only two comparisons occurring between S3 and both R7 and R8 were nonsignificant, though FST was high for both comparisons at 0.71. In addition to the neutral dataset, I also calculated pairwise F_{ST} for the outlier dataset. Interestingly, outlier pairwise F_{ST} was strongly correlated (r= 0.98) with neutral F_{ST}).

Population Structure

An initial assessment of population structure was performed via a PCA using all individuals (Figure 5). The first two axes of the overall PCA explained a cumulative 68% of the variation. The primary separation was observed along PC1 along a south-north gradient. The spread of the ACF and Savannah basins is apparent, as is a lack of overlap. Conversely, the James and Roanoke are tightly

clustered and show little genetic separation. I also performed PCAs separately on northern and southern basins. The north PCA explained 27.1% of the variation in its first two axes. In this more geographically restricted view, ROAN and JAMES showed clear genetic separation. In ROAN, individuals were more tightly grouped (in correspondence with the low pairwise F_{ST} observed in this basin) and ecoregions overlapped in PC space, whereas in JAMES, coastal plain individuals separated from their mountain and Piedmont counterparts along PC1, whereas the Piedmont and mountain showed some overlap. The southern PCA explained 43.3 in its first two axes. It showed separation at the basin level, though slightly less apparent than that found between the James and Roanoke. The Individuals from the ACF separated out along PC1 in a coastal plain to mountain pattern. In the Savannah, individuals from coastal plain and Piedmont sites occupied a similar PC space, whereas the mountain individuals separated out along PC2.

The neutral AMOVA (Table 6) indicated that genetic variance components were significantly greater than zero (p <0.0001) at all spatial scales considered. The largest fraction (67.5%) of variation partitioned among basins, whereas 7.4% was among ecoregions within basins, and only 1.7% was among sites within basins. The remaining 23.4% was nonspatial and partitioned among individuals within sites. For comparison, I also ran an AMOVA in which ecoregions were the highest spatial scale considered (i.e., sites nested within basins, which were nested within ecoregions), but in this AMOVA the ecoregion effect was not significant and actually explained a negative fraction of variance (not shown), indicating that individuals were often more similar to other individuals in a different ecoregion in the same basin than to individuals in the same ecoregion but a different basin.

STRUCTURE was used to determine a best model-supported number of populations (K) and to visualize individual population assignment. At the whole-study scale, the highest mean log-likelihood was found in the K= 6 model. Simpler models with K= 2-5 also showed high mean log-likelihood (especially K= 5), and their individual assignment plots are also shown (Figure 6). The results from this study-wide STRUCTURE analysis show that the primary clustering of individuals corresponded to their location in either the two northern or the two southern basins (i.e., no inferred ancestral populations were shared between the two regions). Within this high-level split, site A1 was the only site wherein no

individuals were admixed. The Savannah mountain sites (S7, S8) also show some uniqueness, though admixture is apparent.

Due to the distinctiveness of the northern versus southern basins in the whole-study analysis, I also performed hierarchical STRUCTURE analysis separately within these regions. In the north, K= 4 was the best supported model (Figure 7). The results of the north models only show support for structuring between basins, with no substantial within-basin structuring by ecoregion or site, though Johns' Creek of the James basin showed unexpected genetic affinity with the Roanoke basin. In the south region (ACF, Savannah), K= 5 was the best supported model(Figure 8). Within the ACF, a pattern of structuring by ecoregion is apparent, especially in the coastal plain where individuals had almost identical ancestral proportions and little admixture. The three mountain sites also had similar ancestral proportions, but with higher levels of admixture with an ancestral population primarily found in the Savanah basin. In the Savannah basin, a similar pattern was found as in the ACF. Here the coastal plain and Piedmont individuals were models as belonging to one primary ancestral population. The Savannah mountains sites were particularly distinct as individuals there assigned predominantly to a unique ancestral population. These findings, along with the low pairwise F_{ST} estimates within ecobasins and the distribution of genetic variation found in the AMOVA, are indicative of hierarchical structuring throughout the studied range. That is, basins drive the highest level of differentiation, with more modest but detectable differences emerging among ecobasins.

Isolation by Distance Versus Environment

At the whole study scale, IBD was apparent (slope= 0.0001275812, R^2= 0.4910087, p= 0.0001), but driven strongly by the large FSTs in comparisons between the northern and southern regions (Figure 9). The degree of IBD varied by basin (Figure 10). Positive and significant IBD slopes were found in the ACF (slope= 0.00013, R^2= 0.49, p= 0.0001) and SAV (slope= 0.00079, R^2= 0.34, p= 0.0001). Contrary to expectation, IBD was not supported in either the James or Roanoke basins. The Roanoke showed a slightly negative relationship (slope= -0.00000895) and was nonsignificant (R^2= 0.00551 , p= 0.6995). A similar lack of positive IBD was apparent in the James basin (slope= -0.000142, R^2= 0.15, p= 0.039).

Models of isolation by distance (IBD), isolation by environment (IBE), and the combined effects of both, were tested against the response variables of neutral and outlier F_{ST} using MRM models (Table 7). I expected that for neutral loci, F_{ST} would be more strongly correlated with stream distance (IBD), whereas for outlier loci, F_{ST} would be strongly correlated with climate and/or physiographic distance (IBE). Contrary to this prediction, I found nearly identical patterns between the two SNP datasets. Importantly, neutral and outlier F_{ST} were strongly positively correlated (r = 0.98), so it makes sense that both genetic distance measures were correlated to environmental distance measures in nearly identical ways. Both models were globally significant, explained a substantial component of the variation in FST (model R^2 was 0.694 and 0.639 for the neutral and outlier data, respectively), and in both models, all three distance measures were significantly associated with FST (all P < 0.05). However, based on standardized slope estimates, which I used as a measure of effect size, the strongest association for both FSTs was a positive relationship with spatial distance, followed by a weaker positive relationship with climate distance, whereas physiographic distance exhibited a weak, negative relationship with both FST. Notably, these are partial tests, which reflect the influence of a given distance measure on FST after the statistical effects of all other distance measures have been accounted for. Thus, test results indicate that, for both neutral and outlier genetic data, even after accounting for the effect of IBD, there is an additional, weaker but significant component of genetic variation accounted for by the two environmental distance measures.

I used a linear mixed model to test whether IBD was steeper in the mountains than the coastal plain, as evidenced by a significantly greater IBD slope, but to control for the effect of barriers on gene flow, I included only the 18 within-basin comparisons with no barriers in this test (Figure 11). One pairwise comparison in the coastal plain (J1 vs. J2) appeared to be an outlier so I conducted the test both with and without this observation. Excluding the outlier, the slope of IBD was significantly higher in mountain (slope= 0.000526, p= 0.0108) and Piedmont (slope= 0.000343, p= 0.00323) comparisons than in the coastal plain. The coastal plain slope wasn't significantly different from zero. Thus, the findings supported my prediction that IBD was stronger in the mountains than the coastal plain. Including the

outlier, resulted in only the Piedmont comparison showing a significant slope (slope= 0.000398, p= 0.0403).

Landscape Influences on Diversity and Differentiation

I conducted random forest (RF) models on both pairwise conditions between sites (response= F_{ST}) and point conditions at sites (response= H_E) in order to elucidate the effect of landscape factors on genetic differentiation and diversity. I determined the best model in each category based on the percent variation explained (R^2) and the contribution of individual predictor variable based on importance scores (%IncMSE). The best-fitting RF model predicting neutral H_E contained 7 predictor variables (including basin and ecoregion as categorical variables), but only explained 8% of the variation in neutral HE (Figure 12, panel A). The two most important predictors were MEANTEMP (importance score= 25) and CATIONEXCHimportance score= 20). Both positively associated with H_E in the model-predicted partial dependence plots (Figure 12, panels B & C). In contrast, partial dependence plots of the other five predictor variables displayed no consistent associations of H_E with drainage area, ecoregion, or distance-to-ocean.

The best fitting RF model including all 465 pairwise comparisons (whole-study scale) explained 72% of variation of pairwise neutral F_{ST}. Only two predictor variables were retained in this model: instream distance and the presence and type of barrier between sites (completely riverine, impoundment, dam, or ocean) (Figure 13, panel A). Instream distance was the most important predictor (importance score= 115) and its partial dependence plot showed a steep increase in model-predicted FST from ~0.14 to ~0.5 over the first 1000 km of spatial distance and a plateau after that (Figure 13, panel B). Barrier type was somewhat less important (importance score= 60). Partial dependence plots indicated that, after accounting for the influence of all other factors, F_{ST} was roughly equivalent (~0.26) in riverine and between-impoundment comparisons, whereas dams between sites increased genetic differentiation slightly (to ~0.29) and oceans (between basin comparisons) represented the greatest barrier to gene flow, increasing FST to ~0.45.

The best-fitting RF model including only same-basin comparisons (n = 105) explained 33% of the variation in neutral F_{ST}. The most important predictors were basin (importance score= 80) stream distance (importance score= 49), and barrier type (importance score= 41) (Figure 14, panel A). Basins followed a pattern of higher F_{ST} among sites in the southern basins and lower F_{ST} in the northern basins (Figure 14, panel B). The modeled relationship between pairwise F_{ST} and instream distance showed a sharp increase after approximately 400 km (Figure 14, panel C). The partial dependence plots of barrier types show that the riverine and impoundment connections influence F_{ST} roughly equally, whereas dams increase it (Figure 14, panel D). By extrapolating the predicted dam influence on F_{ST} onto the instream distance plot, I deduced that, all else being equal, the presence of a dam between sites inflated F_{ST} roughly as much as 450 km of riverine distance.

Outlier Loci Characterization

In order to determine possible functional roles of SNP loci discovered in this study, I ran BLAST searches for all 2585 loci discovered in this study. I then reduced the complexity in interpretation of these results by assigning gene ontology terms to the BLAST hits. In total 429 loci retuned BLAST hits among the full catalog. Of the outlier subset, 44 loci returned BLAST hits and of these 24 were assigned GO terms. A Fisher's exact test indicated that no GO terms were significantly enriched in the outlier loci as compared to all loci. This indicates that no gene functional groups were overrepresented in the outlierloci as compared to all loci. Nonetheless, the most common GO terms (Table 8) included those related to the biological processes (GO:0008150) of metabolism (GO:0008152) and nucleic acid metabolism (GO:0006139, GO:0006259), and to molecular functions (GO:0003674) relating to binding (GO:0003674) and catalytic activity (GO:0005488). In addition, the cellular components (GO:0005575) of cell (GO:0005623) and intracellular (GO:0005622) were frequent GO terms.

Table 3. Results of a principal components analysis used to reduce the dimensionality of environmental variables subsequently used in the redundancy analysis. Column entries show the loadings of each environmental variable onto each of the first two principal components (Comp.1 and Comp.2).

	Comp.1	Comp.2
ELEVMIN	0.538	0.049
GRADIENT	0.328	0.028
MEANTEMP	-0.302	-0.471
TEMPSD	-0.053	0.565
PRECIP	0.291	-0.386
PRECIPCV	0.142	-0.455
CATIONEXCH	0.301	0.183
LCDEVEL	-0.274	0.243
LCFORSHB	0.488	0.083
Percent of Varience (%)	34.4	30.8

Table 4. Estimated genetic diversity measures at each sample site: observed (H_O) and expected (H_E) heterozygosity. Diversity could not be estimated at site S7 because sample size was only one individual.

Site Code	**Basin**	**Ecoregion**	**Site Name**	H_E	H_O
A1	ACF	CP	Aycocks Creek	0.4276	0.4156
A2			Sawhatchee Creek	0.4187	0.4021
A3			Cooleewahee Creek	0.4149	0.4094
A4		PD	Centralhatchee Creek	0.3769	0.3365
A5			New River	0.4130	0.3788
A6		MT	Soque River	0.4191	0.4101
A7			Chattahoochee River	0.4320	0.4279
A8			Chestatee Creek	0.4038	0.3374
S1	Savannah	CP	Beaverdam Creek (A)	0.4220	0.4073
S2			Beaverdam Creak (B)	0.4274	0.4122
S3			Ebenezer Creek	0.4751	0.4478
S4		PD	Horn Creek	0.4231	0.3976
S5			Steven's Creek	0.4204	0.3980
S6		MT	Chauga River	0.4748	0.4383
S7			Chattooga River (A)	na	na
S8			Chattooga River (B)	0.3703	0.3629
R1	Roanoke	CP	Lee's Creek	0.3807	0.3669
R2			Roanoke River side channel	0.3848	0.3810
R3			Quankey Creek	0.3373	0.3361
R4		PD	Goose Creek	0.4032	0.3901
R5			Roanoke Creek	0.3929	0.3839
R6		MT	North Fork Roanoke River	0.4148	0.4002
R7			South Fork Roanoke River	0.4779	0.4825
R8			Tinker Creek	0.4677	0.4755
J1	James	CP	Chickahominy River	0.4201	0.4304
J2			Falling Creek	0.3899	0.3873
J3			Swift Creek	0.3946	0.3910
J4		PD	Hardware River	0.3982	0.4033
J5			Willis River	0.3827	0.3688
J6		MT	John's Creek	0.3969	0.3948
J7			Dunlap Creek	0.4237	0.3964
J8			Calfpasture River	0.4038	0.4008

Table 5. Pair-wise genetic differentiation among all pairs of sites. Weir and Cockerham's F_{ST} is below the diagonal; associated *P*-values are above the diagonal. F_{ST} estimates with $P < 0.05$ are in bold.

	A1	A2	A3	A4	A5	A6	A7	A8	S1	S2	S3	S4	S5	S6	S7	S8	R1	R2	R3	R4	R5	R6	R7	R8	J1	J2	J3	J4	J5	J6	J7	J8
A1		0.5437	0.1977	0.7778	0.1433	0.0229	0.0229	0.0008	0.0228	0.0228	0.0228	0.0229	0.0229	0.0229	1.0000	0.0008	0.0003	0.0005	0.0003	0.0008	0.0007	0.0009	0.0229	0.0229	0.0228	0.0028	0.0028	0.0028	0.0028	0.0028	0.0029	0.0229
A2	-0.005		0.828	0.985	0.8330	0.0229	0.0228	0.0008	0.0228	0.0229	0.0330	0.0229	0.0229	0.0229	1.0000	0.0008	0.0003	0.0005	0.0003	0.0008	0.0008	0.0008	0.0229	0.0229	0.0229	0.0028	0.0029	0.0028	0.0027	0.0028	0.0229	0.0229
A3	0.002	-0.011		0.991	1.0000	0.0229	0.0228	0.0008	0.0228	0.0229	0.0229	0.0229	0.0228	0.0228	1.0000	0.0008	0.0003	0.0005	0.0003	0.0008	0.0008	0.0008	0.0228	0.0227	0.0228	0.0027	0.0028	0.0028	0.0028	0.0028	0.0229	0.0228
A4	-0.005	-0.020	-0.022		0.816	0.0008	0.0016	0.0024	0.0008	0.0008	0.0019	0.0008	0.0008	0.0018	1.0000	0.0007	0.0001	0.0002	0.0001	0.0008	0.0008	0.0008	0.0018	0.0018	0.0007	0.0008	0.0008	0.0008	0.0008	0.0008	0.0008	0.0008
A5	0.005	-0.013	-0.024	-0.006		0.0028	0.0029	0.1738	0.0228	0.0228	0.0229	0.0229	0.0228	0.0228	1.0000	0.0008	0.0003	0.0005	0.0003	0.0008	0.0008	0.0009	0.0229	0.0228	0.0028	0.0028	0.0028	0.0028	0.0028	0.0028	0.0229	0.0228
A6	**0.4009**	**0.3799**	**0.3566**	**0.2755**	**0.2201**		0.0228	0.0008	0.0228	0.0229	0.0228	0.0228	0.1433	0.0229	0.2000	0.0008	0.0003	0.0005	0.0003	0.0008	0.0008	0.0008	0.0229	0.0229	0.0229	0.0028	0.0028	0.0028	0.0028	0.0028	0.0229	0.0229
A7	**0.242**	**0.221**	**0.202**	**0.109**	**0.049**	**0.066**		0.0008	0.0229	0.0229	0.0229	0.0228	0.0228	0.0228	0.2000	0.0008	0.0003	0.0005	0.0003	0.0008	0.0008	0.0008	0.0228	0.0228	0.0229	0.0028	0.0028	0.0028	0.0028	0.0028	0.0228	0.0229
A8	**0.198**	**0.181**	**0.165**	**0.072**	0.022	**0.112**	**0.042**		0.008	0.008	0.018	0.008	0.024	0.018	0.335	0.008	0.001	0.002	0.001	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.007	0.008	0.007	0.008	0.009	0.008
S1	**0.497**	**0.480**	**0.457**	**0.380**	**0.342**	**0.072**	**0.144**	**0.157**		0.541	0.000	0.000	0.000	0.028	0.000	0.008	0.003	0.004	0.003	0.008	0.008	0.008	0.028	0.029	0.028	0.007	0.028	0.028	0.028	0.028	0.029	0.029
S2	**0.465**	**0.449**	**0.426**	**0.345**	**0.302**	**0.074**	**0.131**	**0.134**	0.001		0.600	0.000	0.000	0.029	0.000	0.008	0.003	0.004	0.003	0.008	0.008	0.008	0.028	0.029	0.029	0.008	0.007	0.028	0.028	0.028	0.029	0.028
S3	**0.522**	**0.506**	**0.478**	**0.403**	**0.366**	**0.086**	**0.164**	**0.166**	-0.013	0.001		1.000	0.971	0.099	0.252	0.018	0.008	0.012	0.008	0.018	0.018	0.019	0.099	0.100	0.029	0.018	0.019	0.019	0.018	0.018	0.029	0.028
S4	**0.499**	**0.482**	**0.458**	**0.376**	**0.343**	**0.050**	**0.118**	**0.126**	-0.072	-0.045	-0.079		0.971	0.028	1.000	0.008	0.003	0.005	0.003	0.008	0.008	0.008	0.028	0.029	0.028	0.008	0.008	0.007	0.008	0.008	0.029	0.029
S5	**0.477**	**0.458**	**0.426**	**0.347**	**0.324**	0.044	**0.066**	**0.060**	-0.086	-0.076	-0.080	-0.018		0.028	1.000	0.008	0.003	0.005	0.003	0.008	0.008	0.008	0.028	0.029	0.028	0.007	0.008	0.008	0.008	0.008	0.028	0.028
S6	**0.475**	**0.455**	**0.436**	**0.370**	**0.329**	**0.219**	**0.232**	**0.222**	**0.209**	**0.222**	0.215	**0.177**	**0.103**		1.000	0.017	0.008	0.012	0.008	0.018	0.018	0.018	0.099	0.099	0.029	0.017	0.017	0.017	0.017	0.019	0.029	0.030
S7	0.482	0.464	0.427	0.339	0.301	0.066	0.066	0.112	0.141	0.122	0.158	0.141	0.146	0.161		1.000	0.125	0.142	0.124	1.000	0.169	1.000	1.000	1.000	0.199	0.166	0.168	0.166	0.167	1.000	1.000	1.000

S8	0.569	0.551	0.542	0.491	0.464	0.414	0.402	0.386	0.419	0.426	0.432	0.391	0.334	0.104	0.390		0.001	0.002	0.001	0.008	0.008	0.008	0.018	0.018	0.008	0.008	0.008	0.008	0.007	0.008	0.008
R1	0.753	0.744	0.737	0.706	0.697	0.684	0.673	0.653	0.695	0.698	0.713	0.683	0.652	0.528	0.711	0.488		0.894	0.342	0.001	0.213	0.004	0.466	0.193	0.025	0.004	0.007	0.001	0.001	0.001	0.003
R2	0.758	0.748	0.741	0.708	0.700	0.685	0.671	0.650	0.696	0.698	0.716	0.685	0.656	0.526	0.718	0.487	-0.012		0.719	0.002	0.469	0.019	0.489	0.107	0.009	0.004	0.005	0.002	0.002	0.002	0.005
R3	0.733	0.723	0.718	0.685	0.673	0.665	0.653	0.633	0.675	0.678	0.690	0.661	0.624	0.505	0.680	0.469	-0.001	-0.008		0.002	0.229	0.047	0.766	0.349	0.550	0.177	0.146	0.001	0.002	0.001	0.003
R4	0.777	0.768	0.759	0.727	0.722	0.700	0.685	0.664	0.713	0.714	0.733	0.704	0.684	0.552	0.742	0.520	0.032	0.030	0.036		0.008	0.007	0.054	0.072	0.008	0.008	0.007	0.008	0.008	0.008	0.007
R5	0.745	0.735	0.727	0.693	0.681	0.667	0.653	0.633	0.680	0.681	0.697	0.665	0.634	0.497	0.690	0.468	0.002	-0.005	0.001	0.045		0.016	0.108	0.055	0.024	0.008	0.009	0.008	0.008	0.008	0.008
R6	0.784	0.775	0.764	0.732	0.731	0.704	0.686	0.665	0.717	0.717	0.740	0.711	0.695	0.559	0.759	0.523	0.024	0.013	0.008	0.054	0.018		0.018	0.019	0.008	0.007	0.008	0.008	0.008	0.008	0.008
R7	0.768	0.758	0.744	0.712	0.709	0.672	0.651	0.632	0.689	0.689	0.712	0.683	0.671	0.507	0.723	0.485	-0.003	-0.008	-0.014	0.020	0.007	0.017		0.101	0.029	0.018	0.018	0.018	0.017	0.018	0.029
R8	0.767	0.757	0.745	0.712	0.707	0.675	0.655	0.637	0.691	0.691	0.712	0.683	0.667	0.509	0.716	0.489	-0.003	-0.004	-0.003	0.024	0.013	0.026	0.019		0.029	0.018	0.018	0.018	0.016	0.018	0.029
J1	0.759	0.749	0.731	0.694	0.703	0.648	0.621	0.599	0.670	0.664	0.702	0.674	0.677	0.478	0.753	0.452	0.032	0.036	-0.007	0.083	0.036	0.133	0.115	0.075		0.008	0.007	0.937	1.000	1.000	0.630	1.000
J2	0.764	0.754	0.738	0.704	0.709	0.668	0.645	0.623	0.688	0.683	0.716	0.688	0.684	0.504	0.752	0.470	0.049	0.039	-0.003	0.074	0.043	0.119	0.087	0.065	0.082		0.008	1.000	1.000	1.000	0.992	1.000
J3	0.773	0.763	0.748	0.714	0.720	0.678	0.655	0.632	0.696	0.692	0.725	0.697	0.695	0.520	0.766	0.484	0.041	0.044	-0.005	0.079	0.036	0.113	0.098	0.078	-0.115	-0.046		1.000	1.000	1.000	0.976	0.992
J4	0.787	0.779	0.770	0.738	0.733	0.711	0.696	0.675	0.724	0.724	0.746	0.716	0.696	0.570	0.762	0.538	0.169	0.171	0.158	0.182	0.189	0.203	0.186	0.188	-0.035	-0.065	-0.042		0.016	0.008	0.008	0.039
J5	0.773	0.764	0.754	0.721	0.716	0.692	0.677	0.657	0.706	0.706	0.727	0.698	0.679	0.546	0.741	0.515	0.143	0.133	0.128	0.162	0.151	0.181	0.152	0.144	-0.071	-0.085	-0.060	0.036		0.008	0.009	0.254

J6	**0.777**	**0.768**	**0.760**	**0.727**	**0.721**	**0.702**	**0.687**	**0.667**	**0.715**	**0.716**	**0.734**	**0.706**	**0.685**	**0.556**	0.741	**0.525**	**0.069**	**0.067**	**0.068**	**0.078**	**0.096**	**0.087**	**0.062**	**0.065**	-0.064	-0.072	-0.069	**0.063**	**0.041**		0.008	0.007
J7	**0.781**	**0.772**	**0.761**	**0.728**	**0.724**	**0.695**	**0.678**	**0.657**	**0.709**	**0.709**	**0.731**	**0.703**	**0.688**	**0.547**	0.749	**0.525**	**0.138**	**0.132**	**0.125**	**0.149**	**0.158**	**0.180**	**0.153**	**0.151**	-0.013	-0.047	-0.025	**0.038**	**0.031**	**0.061**		0.027
J8	**0.769**	**0.760**	**0.749**	**0.716**	**0.712**	**0.682**	**0.664**	**0.643**	**0.699**	**0.698**	**0.720**	**0.693**	**0.678**	**0.530**	0.737	**0.508**	**0.138**	**0.137**	**0.127**	**0.153**	**0.152**	**0.183**	**0.147**	**0.138**	-0.050	-0.077	-0.050	**0.012**	0.001	**0.051**	**0.033**	

Table 6. Results of an analysis of molecular variance conducted on the neutral dataset. *P*-values are based on 10,000 permutations.

Level	Degrees of freedom	Molecular variance	Percent of variation	*P*
Among Basins	3	514.1	67.5	<0.0001
Among Ecoregions Within Basins	8	56.1	7.4	<0.0001
Among Sites Within Ecoregions	19	13.1	1.7	<0.0001
Among Individuals Within Sites	108	178.4	23.4	
Total	138	761.7504	100	

Table 7. Results of two matrix regression models comparing either neutral and outlier pairwise F_{ST} to a series of spatial and environmental distance measures.

Response	Model R^2	Predictor	Standardized estimate	*P*
Neutral *FST*	0.694	Intercept	0.29843	0.0001
		Stream Distance	0.1063	0.0001
		Climate Distance	0.06381	0.0001
		Physiographic Distance	-0.01282	0.0003
Outlier *FST*	0.639	Intercept	0.492	0.0001
		Stream Distance	0.130137	0.0001
		Climate Distance	0.0586	0.0001
		Physiographic Distance	-0.00933	0.0183

Table 8. Counts of GO-slim gene ontology (GO) terms among outlier loci.

GO Class ID	Definitions	Counts	GO Class ID	Definitions	Counts	GO Class ID	Definitions	Counts
GO:0003674	**molecular_function**	23	GO:0008150	**biological_process**	21	GO:0005575	**cellular_component**	16
GO:0005488	binding	12	GO:0008152	metabolism	13	GO:0005623	cell	13
GO:0003824	catalytic activity	9	GO:0006139	nucleobase, nucleoside, nucleotide and nucleic acid metabolism	4	GO:0005622	intracellular	10
GO:0005515	protein binding	4	GO:0016787	hydrolase activity	3	GO:0005856	cytoskeleton	4
GO:0016740	transferase activity	4	GO:0019538	protein metabolism	3	GO:0005634	nucleus	4
GO:0003676	nucleic acid binding	4	GO:0006259	DNA metabolism	2	GO:0005635	nuclear membrane	3
GO:0008092	cytoskeletal protein binding	2	GO:0009058	biosynthesis	2	GO:0005737	cytoplasm	2
GO:0003677	DNA binding	2	GO:0016043	cell organization and biogenesis	1	GO:0005929	cilium	2
GO:0000166	nucleotide binding	2	GO:0009607	response to biotic stimulus	1	GO:0005815	microtubule organizing center	1
GO:0003779	actin binding	2	GO:0009605	response to external stimulus	1	GO:0005739	mitochondrion	1
GO:0003774	motor activity	1	GO:0007165	signal transduction	1			
GO:0003700	transcription factor activity	1	GO:0007154	cell communication	1			
GO:0005215	transporter activity	1	GO:0006091	generation of precursor metabolites and energy	1			
GO:0003723	RNA binding	1	GO:0006950	response to stress	1			
			GO:0008219	cell death	1			
			GO:0006464	protein modification	1			

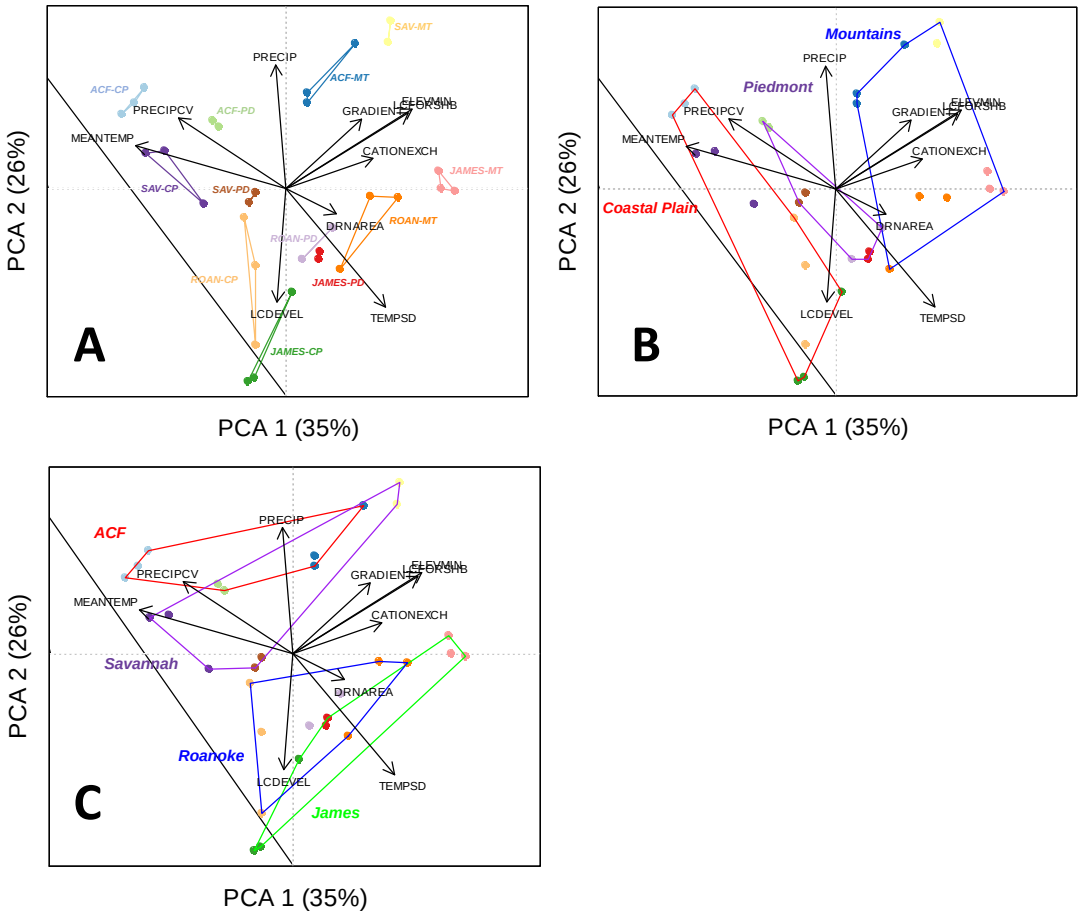

Figure 2. Ordinations showing the results of a principal components analysis on nine environmental (habitat) variables measured for each study site. Each point is a site, whereas vectors illustrate the loading of each environmental variable onto the first two components. Polygons enclose either A) Sites in the same ecobasin, B) Sites in the same ecoregion group, or C) Sites in the same basin.

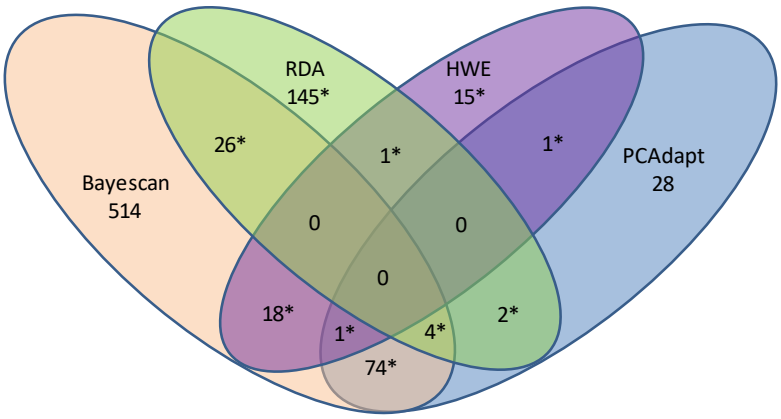

Figure 3. Venn diagram showing the distribution and overlap of outlier SNP loci among four different detection methods [Bayescan, redundancy analysis (RDA), Hardy-Weinberg equilibrium (HWE) tests, and PCAdapt]. Both the northern and southern RDAs are represented by the one bubble. Counts with asterisk indicate those loci that were included in the outlier dataset.

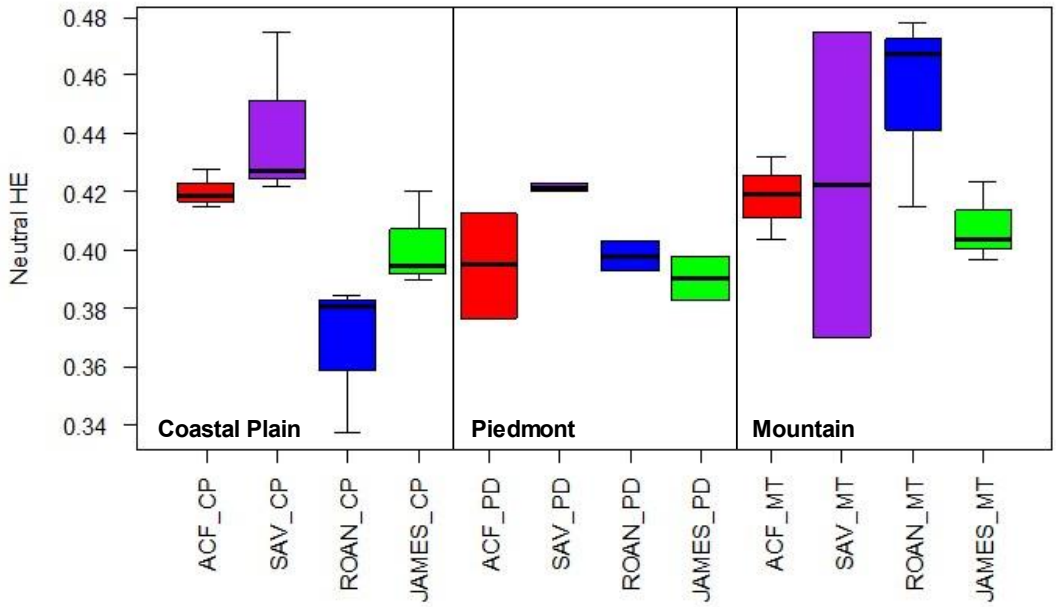

Figure 4. Boxplots showing the distribution of neutral expected heterozygosity (i.e., genetic diversity; H_E). Boxes are grouped by ecoregion and color-coded by basin. The only apparent trend is an increase in diversity between coastal plain and mountain sites in the Roanoke basin.

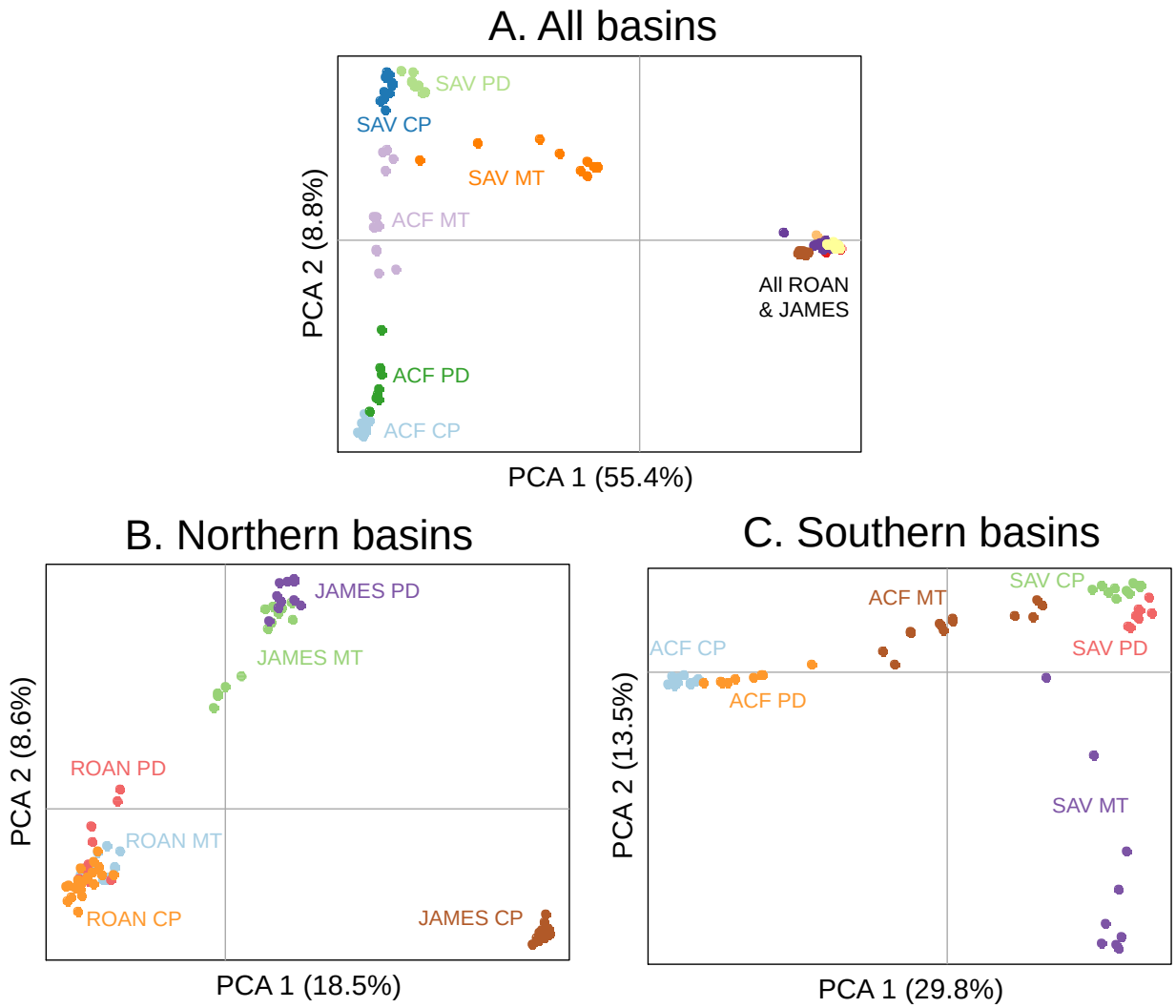

Figure 5. Ordinations showing the results of principal components analyses (PCAs) based on allele frequencies. Each point is an individual. For clarity, I show results for PCAs run separately at three different spatial extents: A) All basins together, B) Northern basins (Roanoke and James) only, and C) Southern basins (Savannah and ACF) only.

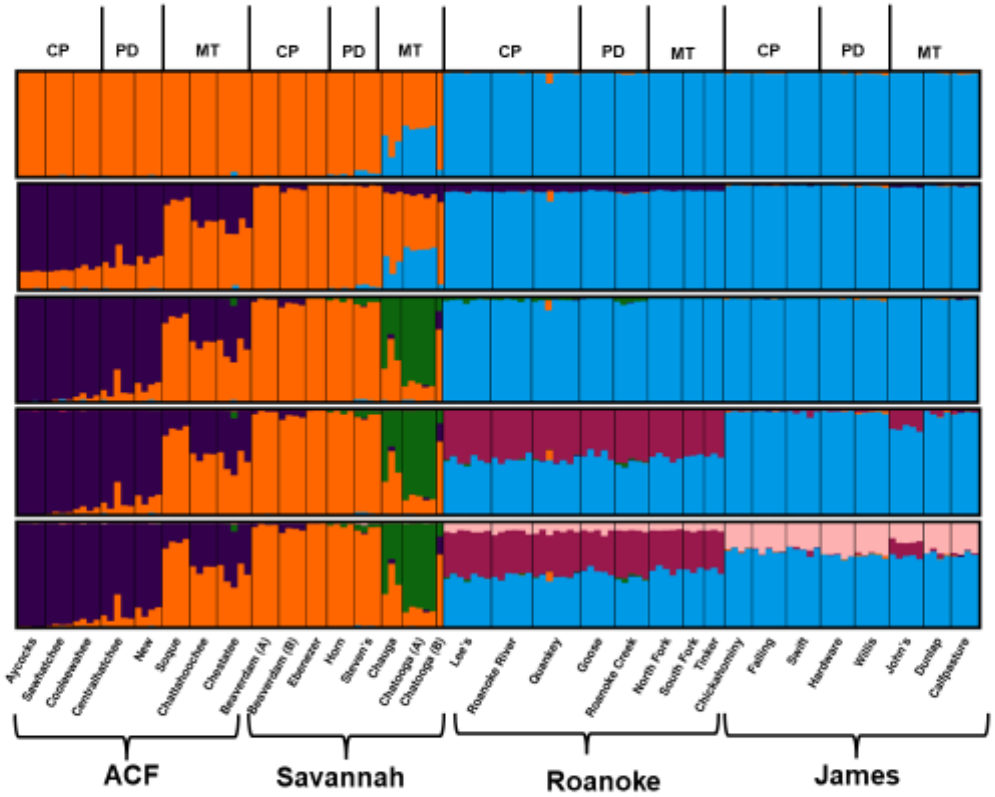

Figure 6. Results of the STRUCTURE analysis of all individuals (all four basins). Shown are the results of the best-fitting models featuring *K*= 2 through 6 (top to bottom). Each vertical bar represents one individual, and colors indicate the proportional assignment of each individual's ancestry to each of *K* inferred genetic clusters. Brackets indicate basins.

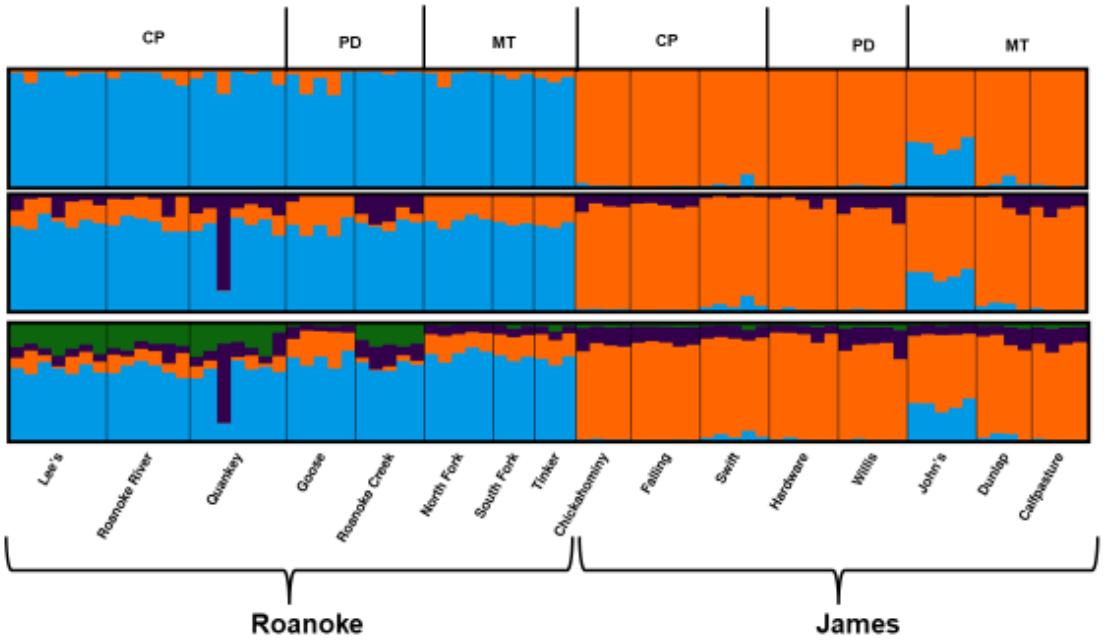

Figure 7. Results of the STRUCTURE analysis of all individuals from the northern region (Roanoke and James basins). Shown are the results of the best-fitting models featuring *K*= 2 through 4 (top to bottom). Each vertical bar represents one individual, and colors indicate the proportional assignment of each individual's ancestry to each of *K* inferred genetic clusters. Brackets indicate basins.

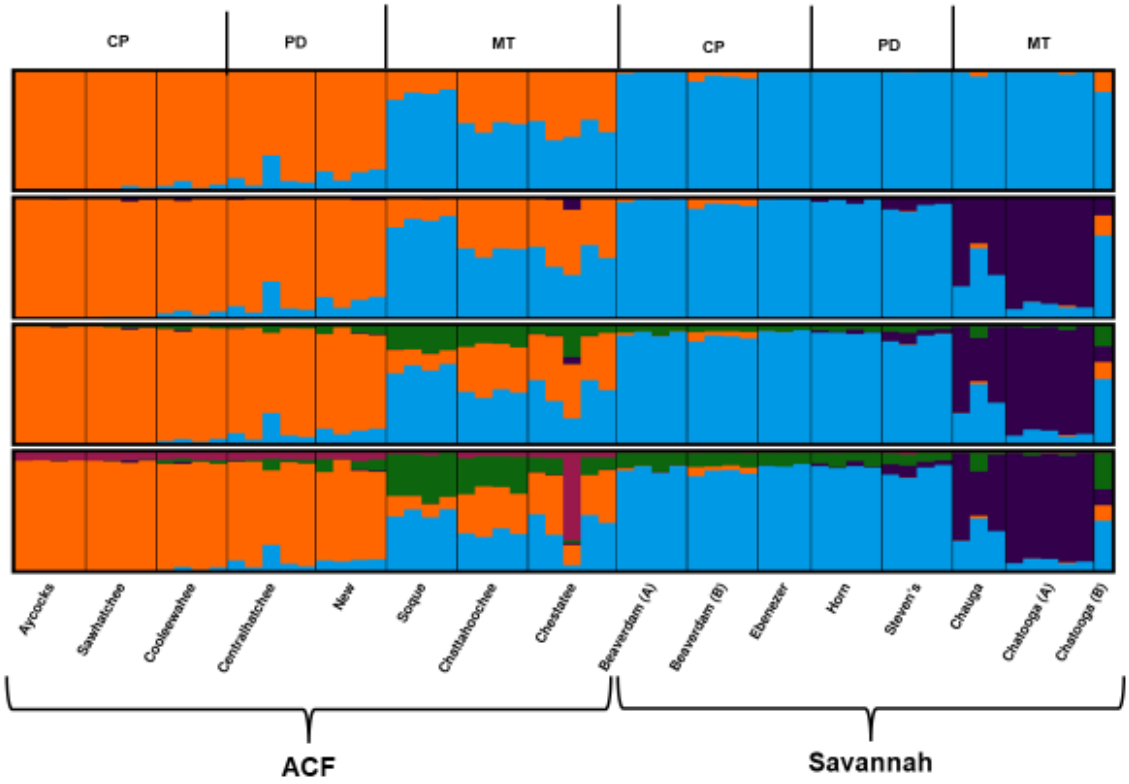

Figure 8. Results of the STRUCTURE analysis of all individuals from the southern region (ACF and Savannah basins). Shown are the results of the best-fitting models featuring *K*= 2 through 5 (top to bottom). Each vertical bar represents one individual, and colors indicate the proportional assignment of each individual's ancestry to each of *K* inferred genetic clusters. Brackets indicate basins.

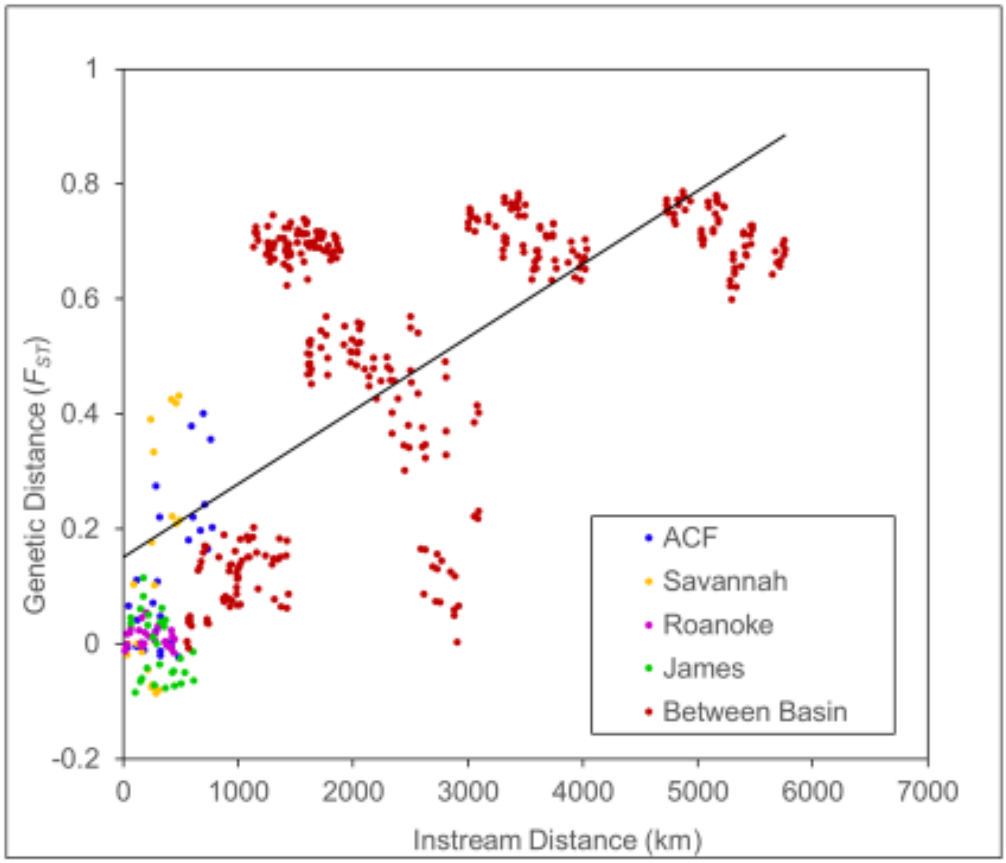

Figure 9. The relationship between along-the-stream spatial distance (km) and genetic distance (F_{ST}) at full study extent. Each point is a pairwise comparison of two sites, which are grouped into within-basin or between-basin categories. A linear trend line is fit to the relationship, as determined by MRM models.

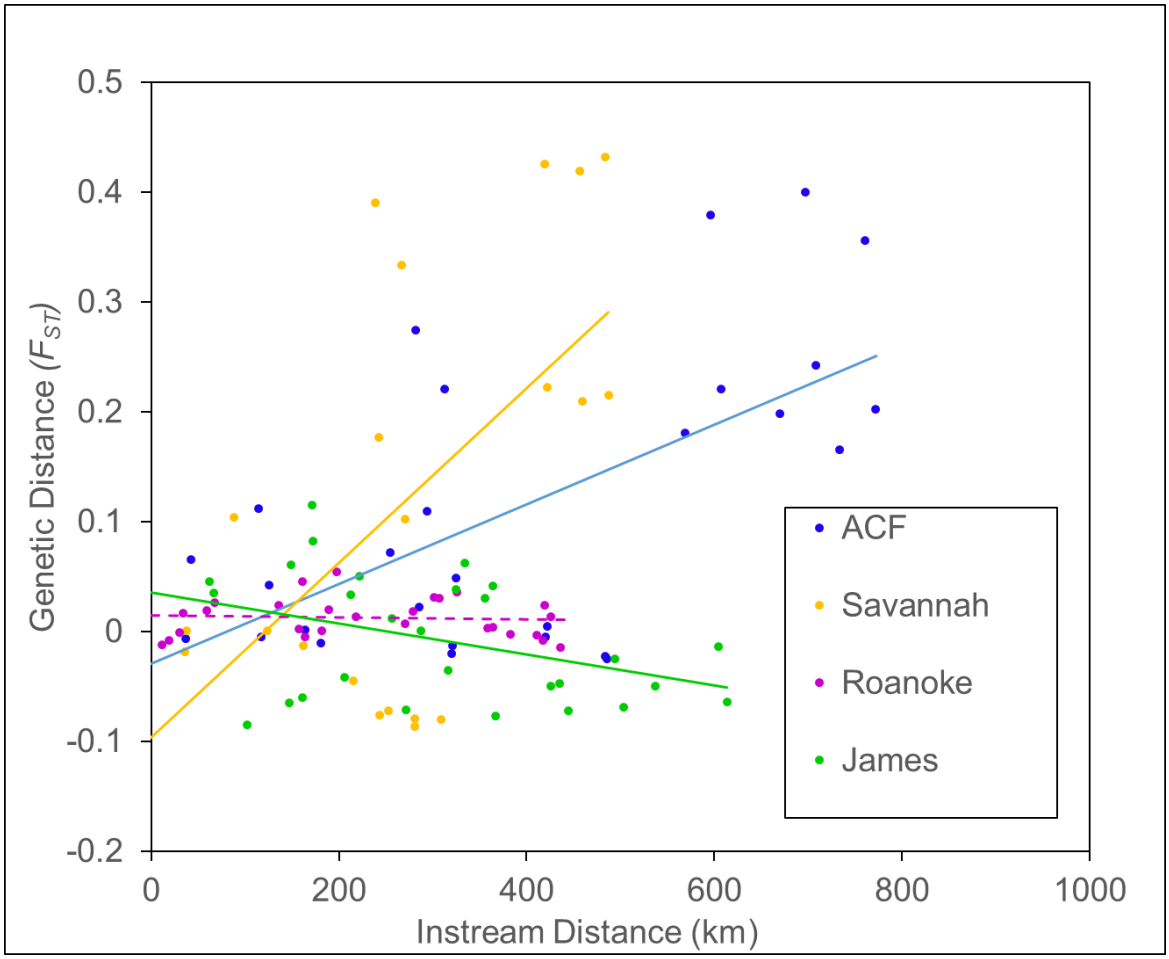

Figure 10. The relationship between along-the-stream spatial distance (km) and genetic distance (F_{ST}), excluding between-basin comparisons. Each point is a pairwise comparison of two sites, which are grouped by basin. Linear trend lines are fit to these relationships, as determined by MRM models. The dotted line fit to the Roanoke points did not differ significantly from zero; all other slopes were significantly different from zero.

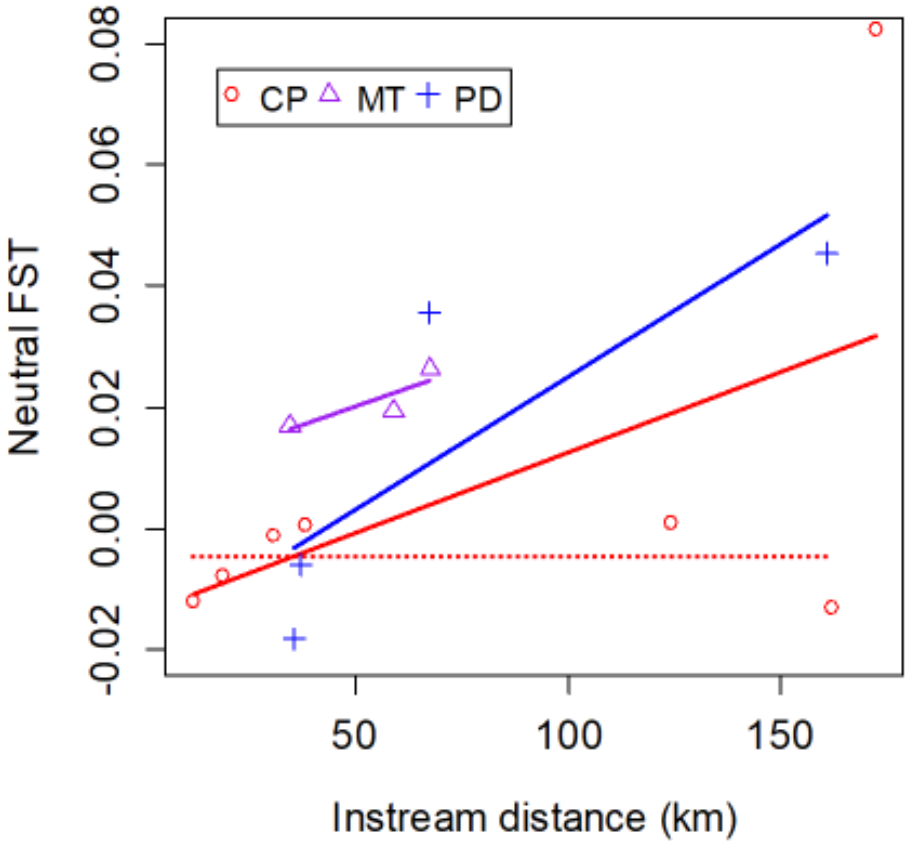

Figure 11. The relationship between along-the-stream spatial distance (km) and genetic distance (F_{ST}), only for sites within the same basin and not separated by any type of barrier (dam or reservoir). Each point is a pairwise comparison of two sites, which are grouped by ecoregion. Linear trend lines are fit to these relationships, as determined by a linear regression model. Based on this model, all three slopes (solid lines) were significantly greater than zero. However, when the top right point from the coastal plain, a potential outlier, was removed, the slope for the coastal plain (dotted red line) was not significantly different from zero.

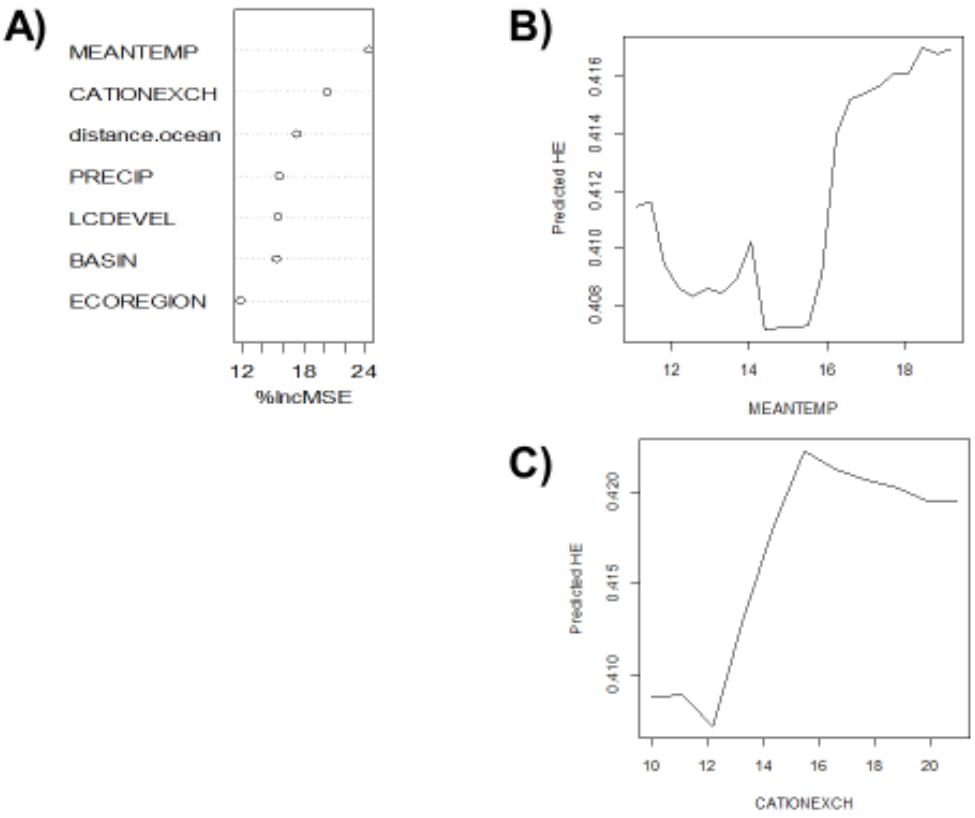

Figure 12. Results from the best-fitting random forest model predicting genetic diversity (H_E) at sites. A) Importance scores for the 7 variables included in the model. B) Partial dependence plot showing the relationship between mean annual air temperature and predicted H_E. C) Partial dependence plot showing the relationship between cation exchange capacity and predicted H_E.

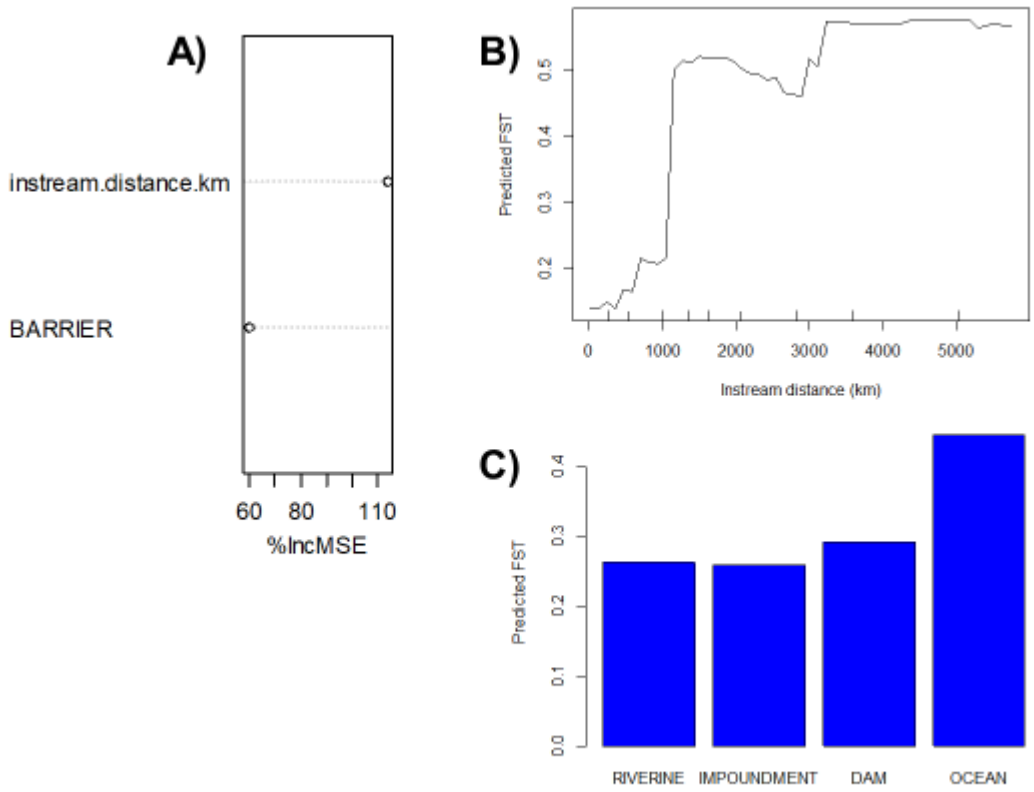

Figure 13. Results from the best-fitting random forest model predicting pairwise F_{ST} between sites across the whole study area. A) Importance scores for the 2 variables included in the model. B) Partial dependence plot showing the relationship between instream distance (km) and predicted F_{ST}. C) Partial dependence plot showing the relationship between barrier conditions between pairs of sites (free-flowing riverine connection versus various types of barriers) and predicted F_{ST}.

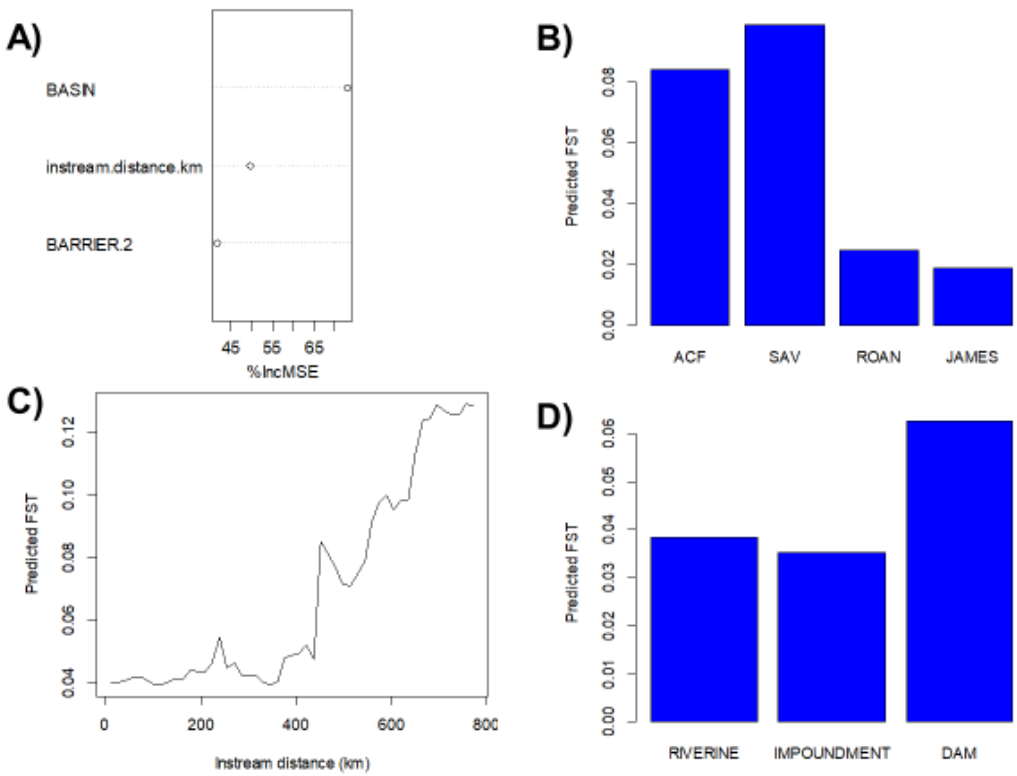

Figure 14. Results from the best-fitting random forest model predicting pairwise F_{ST} between sites, based only on comparisons within the same basin. A) Importance scores for the 3 variables included in the model. B) Partial dependence plot showing the relationship between basin and predicted F_{ST}. C) Partial dependence plot showing the relationship between instream distance (km) and predicted F_{ST}. D) Partial dependence plot showing the relationship between barrier conditions between pairs of sites (free-flowing riverine connection versus impoundment versus dam) and predicted F_{ST}.

CHAPTER 4: DISCUSSION

Novelty of this Research

To my knowledge, mine is the first published study to use population genomics approaches to study the distribution of genetic variation in a non-salmonid freshwater fish across multiple ecoregions and drainage basins of the Atlantic slope. In addition, few studies of fishes in this region have focused on contemporary eco-evolutionary patterns in widespread generalist species such as RBS, but rather have focused primarily on phylogenetic or conservation-genetic questions in more-popular gamefishes or narrowly distributed imperiled species. My focus on RBS allowed me to capture processes playing out across diverse landscapes (e.g., mountain versus coastal plain ecosystems) in a relatively abundant species whose genomic diversity has not been persistently bottlenecked by anthropogenic impacts, which could obscure the signals of gene flow and local adaptation. As such, the organisms and modeling approaches used herein have revealed generalizable results as to what shapes the distribution of genetic diversity on the landscape.

Ecobasins as Unique Environmental Replicates

Contrary to my expectations, ecoregions were not environmentally homogeneous across basins. Although physiographic variables showed consistent variation across ecoregion groups (i.e., higher elevation and gradient and more forest in the mountains, the opposite being true in the coastal plain, and the Piedmont being intermediate), climate variables varied more among basins according to their latitudinal position than among ecoregions within basins (i.e., colder, more-variable air temperatures and lower, less-variable precipitation amounts in northern than southern basins). Although this indicates that my initial experimental design was unsuccessful at creating homogeneous groups within a factorial design (4 basins, 3 ecoregion groups), it does indicate that my sites sampled clear and meaningful gradients in climate and physiographic conditions I posited would be important for the demographics and evolution of RBS. Fortunately, I was able to employ some gene-environment-association analyses that did not depend on replication of sites within discrete, homogeneous groups (e.g., RDAs, MRMs). The wide range of

climate and physiographic conditions I observed both within and between basins illustrates two additional points. First, it illustrates the high adaptability of RBS to diverse evolutionary environments, highlighting the potential utility of this species as a model for understanding the genomic architecture behind local adaptation in general, and potential adaptation to climate change in particular. Second, it illustrates that ecobasins are likely to be a useful organizing spatial framework for studying evolutionary processes, as this finer-grained perspective seems to explain more variation in fish biodiversity than ecoregions or basins alone (Frimpong and Angermeier 2010).

Hierarchical Population Structure

As hypothesized, RBS showed hierarchical population structuring, whereby basin boundaries corresponded to the greatest amount of genetic differentiation, with progressively less pronounced genetic distinctions occurring among ecobasins within the same basin and among sites within the same ecobasin. This was supported by AMOVA, PCA and STRUCTURE results. Strong basin-level differentiation comes as no surprise, given that the populations of RBS I sampled in different basins were separated by completely inhospitable marine habitat, various natural (waterfalls) and manmade (dams) barriers, a minimum instream distance of 553 km (mean = 2623 km), and presumably many generations of reproductive isolation. Natural historical dispersal between the river-mouths of the northern basins theoretically could have occurred during times of lowered sea level prior to the Holocene, or among adjacent coastal plain streams during prehistoric flooding events (Jenkins and Burkhead 1994), but neither of these dispersal mechanisms could have connected the Savannah and ACF basins in the southern part of the study area, whose lower courses are separated by a large geographic distance and many intervening river basins and whose river mouths are separated by the Florida peninsula. These formidable barriers between the southern basins could explain the generally greater differentiation between the Savannah and ACF than between the Roanoke and James basins. Other potential natural, historical dispersal mechanisms that could have connected basins within both of these regions are headwater piracy/stream capture and subterranean dispersal through in limestone-rich areas. There is geologic and faunal evidence for stream capture and subterranean dispersal in the mountains of the Roanoke and James basins (Jenkins

and Burkhead 1994) and the ACF-Savannah (Dubose 2017). However, if such events and resulting fish dispersal were recent enough to strongly influence genetic patterns, we would expect inter-basin mountain comparisons to show lower genetic differentiation than inter-basin coastal plain comparisons. Although this appears to be the case in the whole-study-area PCA plot (Figure 5a), inter-basin mountain relatedness is not apparent in PCA plots subdivided by region (Figure 5b and 5c), IBD plots, or STRUCTURE plots. As such, if headwater piracy (or other prehistoric dispersal mechanisms) have played a role in inter-basin dispersal for RBS, it is no longer evident in the genetic data.

Whereas strong basin-scale population structure was expected, sub-basin structuring by ecoregion was unexpectedly weak. However, this was in part due to differences between the northern and southern regions. The two northern basins showed a distinct lack of IBD, formed no STRUCTURE clusters corresponding to ecobasin, and the Roanoke basin in particular showed little separation of ecobasins in PCA space. The James basin exhibited more discernible ecobasin separation in PCA space, but primarily an expectedly large difference between the coastal plain and the other two ecobasins. By contrast, the two southern basins showed strong IBD, expected longitudinal separation of ecobasins in PCA space, and some evidence for mountain versus non-mountain genetic clustering in STRUCTURE. Several factors could explain why these basins showed this trend. One possible explanation could be that RBS have been more frequently moved among ecoregions by humans in northern than southern basins. RBS is a popular sportfish, and although I am aware of no records of state or federal agencies intentionally transporting and stocking RBS, historical recordkeeping on such practices is generally poor (Jenkins and Burkhead 1994). Moreover, given that the northern basins are more urbanized than the southern basins according to the environmental PCA, it seems that private citizens would have been more likely to transport and introduce RBS in the north than the south. Unfortunately, I lack divergence time estimates for RBS populations, so I cannot directly evaluate the likelihood of recent (20th century) introductions. However, I find it unlikely that such introductions fully explain weak genetic differentiation in the northern basins, for two reasons. First, weak differentiation was a widespread phenomenon, whereas under a hypothesis of anthropogenic dispersal, I would predict unexpectedly high relatedness to be localized in certain streams and watersheds

(i.e., those that donated or received propagules). Second, in a companion morphological study of these same individual fish, Walker (2019) found that RBS morphological attributes (e.g., eye diameter, head roundedness, caudal peduncle length) varied significantly and consistently among ecoregions in the Roanoke and James basins. Assuming morphology is at least partly under genetic control, we would expect such local adaptations – presumably for local hydrological conditions – to have been homogenized if fish were being anthropogenically moved among ecoregions. A second potential explanation is that Pleistocene era sea level changes and glacial expansions may have left northern basin RBS in small refugia populations that later spread throughout after glacial retreat. A third potential explanation can be found in the river morphology of these basins, where the northern basins show weaker elevation gradients compared to the southern basins, leading to less restriction for gene flow.

Landscape Determinants of Genetic Diversity and Gene Flow

Genetic diversity enhances individual fitness, and at the population scale provides the basis for adaptation via selection, persistence in changing environments, and ultimately is the engine for the creation of new biodiversity. Gene flow can pass novel alleles and adaptations from population to population and helps maintain a species' overall genetic diversity, but gene flow is constrained by the dispersal limitations of a species and the distribution of dispersal barriers across the landscape. From a practical standpoint, examining patterns of variation at neutral genetic loci allows us to infer how the environment modulates genetic drift and gene flow, and thereby infer contemporary demographic and evolutionary processes, whereas examining patterns of variation at loci under selection allows us to infer local adaptation in response to the environment. I set about this thesis with both goals in mind, in the hopes of identifying two complimentary sets of SNP loci that were useful for answering either neutral or adaptive questions. Unfortunately, the SNPs I identified as outliers potentially under selection showed little resolving power beyond that provided by neutral SNPs (see next section), limiting my ability to answer questions about how the landscape promotes local adaptation. As such, I focus the remainder of this section on questions answerable by the selectively neutral SNPs I identified, especially in answering 1) how diversity varies longitudinally (i.e., with ecoregion and stream size), 2) whether gene flow is

generally higher in coastal plain and lower in mountain ecoregions, and 3) which landscape attributes best predict genetic diversity within populations and gene flow between populations.

Previous studies have found that genetic diversity increases downstream and hypothesized that this was due to the effect of downstream-directed rivers flows impeding upstream dispersal (Paz-Vinas, Loot et al. 2015). I hypothesized that this would be true of RBS because they are not known to make migrations upstream or to be of a particularly high dispersal potential. Contrary to my prediction, there was no apparent relationship between gene diversity [indexed by expected heterozygosity (H_E) at neutral loci and any of three surrogates for longitudinal position: (1) ecoregion, (2) distance-from-ocean, or (3) watershed area. This may indicate that dispersal occurs at rates roughly equal both upstream and downstream. Rather than longitudinal position, ecoregion, or stream size, random forest models indicated that H_E was best predicted by the mean annual air temperature and cation exchange capacity of sites, both being positively related to H_E. The temperature effect corresponds with the thermal optimum for RBS and suggest that RBS populations in warmer waters experience faster growth relative to cooler waters and thus may sustain larger effective population sizes. Similarly, sites with greater cation exchange capacity likely exhibit greater groundwater inputs, which may modulate discharge variation and allow more constant population sizes. However, neither of these effects explained a wide range in H_E, and the RF model for H_E explained relatively little variation (~8%), so these results should not be overemphasized.

Unlike genetic diversity, my observations with regard to F_{ST} were generally consistent with my predictions about how various environmental attributes would influence gene flow. Namely, because centrarchid sunfishes generally exhibit strong fidelity to localized territories (Gerking 1953; Gatz and Adams 1994), I predicted that RBS would show distance-mediated gene flow and thus strong IBD. Based on IBD scatterplots, this was evident in the Savannah and ACF basins, whereas IBD was virtually nonexistent in the Roanoke and James basins, potentially for reasons described in the previous section. Notably, the influence of distance on F_{ST} typically is confounded with the influence of barriers, because the farther two populations are located from each other, the more likely they are to be separated by a barrier. Random forest models allowed me to parse these two influences and measure their typical effects

on F_{ST} while accounting for each other. The global model explained a large percentage of the variation (~72%) and indicated that basin boundaries (i.e., ocean barriers) profoundly inflated F_{ST}, overwhelming the signals of other barriers or IBD. Removing this basin-barrier effect and looking only at same-basin comparisons, I found that dams had a large effect on gene flow: the presence of a dam between sites inflated F_{ST} approximately the same amount as 450 kilometers of completely riverine distance. This sobering result illustrates how quickly anthropogenic barriers like dams may be artificially driving genetic divergence of aquatic populations, through a combination of reduced gene flow and increased drift, the ultimate evolutionary consequences of which are yet unknown. This is similar to the 1200 km determined for Roanoke Logperch (*Percina rex*) by Roberts et al. (2013) and may indicate the degree to which dispersal-capacity can influence gene flow across barriers. Also, my findings here correspond with the 378 km determined for the co-occurring centrarchid, Roanoke bass (*Ambloplites cavifrons*) (Eschenroeder and Roberts 2020), and may show that the larger *A. cavifrons* is a slightly better disperser than RBS. Unlike dams, site comparisons separated by the upper reaches of impoundments (but not a dam) did not show any inflation of F_{ST} relative to strictly riverine conditions. Although RBS are more commonly found in riverine than lacustrine environments, it makes sense that impounded sections of river would not form a strong movement barrier for a sunfish like RBS, as they do for benthic fishes like darters (Roberts et al. 2013). The RF model also allowed me to control for the effect of basin membership when estimating these relationships, and the strong influence of basin membership on estimated mean F_{ST} (higher in southern than northern basins) indicates the importance of doing so in future investigations of this kind. Given the strong effect of dams on gene flow, I used only those site-comparisons with strictly riverine connections to answer the final question of this part of the thesis: Is gene flow higher in coastal plain than mountain ecoregions? I predicted it would be, given that upland streams generally have higher gradient and more natural movement barriers like waterfalls, riffles, and shoals, which should resist movement for a laterally compressed fish like RBS. A linear model testing for an interaction between ecoregion group and IBD showed mixed support for this idea. When all 15 applicable pairwise comparisons were included, the IBD slope was significantly greater than zero in all three ecoregion groups, but the

interaction was not significant. However, when one potential outlier observation from the James coastal plain (Chickahominy River x Falling River) was excluded, the coastal plain IBD slope became flat and was significantly shallower than that for the Piedmont or mountains. The spurious comparison between Chickahominy and the Falling River may be due to the latter's location in the urban center of Richmond, VA, where smaller dams, culverts, and channel modifications could certainly hinder dispersal. At any rate, due to limited sample size of barrier-free comparisons, the power of this test is limited. A finer-grained assessment of segment-scale gene flow across ecoregions, with larger sample sizes, would help address the question of how and whether movement resistance varies with ecoregion.

The Search for Adaptive Loci

Altogether, 287 SNP loci were discovered that correlated in the RDA more strongly with measured environmental conditions than with the predominant pattern of population structure, behaved as FST outliers in one or more other tests, or both, and were categorized as putatively adaptive loci. Unfortunately, subsequent analyses showed that genetic variation (F_{ST}) at these outlier loci was very strongly correlated (r = 0.98) with F_{ST} based on the neutral dataset, and like neutral F_{ST}, outlier F_{ST} was better predicted by IBD than by IBE. Thus, the outlier loci showed no evidence for a detectable association with the measured environmental attributes and therefore provided no additional information on evolution beyond that provided by the neutral loci. Interestingly, my search of the literature revealed no other tests for such correlations between neutral and outlier loci, with one exception. Candy et al. (2015) compared pairwise F_{ST} and H_E between putatively neutral and outlier datasets and found that the mean F_{ST} was higher for the outlier dataset and advocated for the greater utility of the outlier SNPs in identifying stocks of Eulachon (*Thaleichthys pacificus*), an anadromous smelt occurring over a range of environmental conditions in the Pacific ocean. My mean and range of F_{ST} (outlier mean: 0.48 ; neutral mean: 0.42) did not differ as drastically as did the authors' but did show a trend of greater differentiation in the outlier SNPs. Whether this constitutes a general phenomenon for all such population genomics studies or a unique case for certain datasets remains unclear. Hopefully more studies will report on this in the future.

Despite the lack of association of outlier SNPs with environmental variables, I used BLAST to assess whether these loci showed greater association than neutral loci with functional genes. Again, the exact test did not detect significant enrichment in gene ontology terms associated with the outlier loci. However, some outlier SNPs did map closely to gene functions associated with metabolism and nucleic acid metabolism, suggesting that at least some of these outlier loci may be located on or near function-coding regions and that some "true" adaptive genetic variation was captured by a handful of the SNPs in this study. Perhaps the "signal" of this variation was overwhelmed by the "noise" from other outlier SNPs that exhibited minimal or opposite responses to environmental variation. Further ground-truthing of these loci was beyond the scope of this study but it provided a distinct angle by which to view the RBS genome. If GEA and outlier-test analyses did not truly detect a signature of local adaptation, several alternative hypotheses may be supported. First, phenotypic plasticity accounts for the majority of redbreast sunfish fitness in a variety of environments. This would indicate variation in gene expression patterns in genes relevant to variable environments. I suspect that environmentally driven gene expression variation does exist alongside sequence variation. A common-garden experiment would ultimately be required to elucidate the relative roles of phenotypic plasticity and genetic differentiation. Conversely, phenotypic plasticity may have even promoted local adaptation in redbreast sunfish over long-term time scales as it may allow individuals to cope with novel selection pressures, thus allowing adaptive mutations and pre-existing adaptive alleles to accumulate and spread through populations (Levis & Pfennig, 2016). Additionally, the ability to have a plastic phenotype is ultimately controlled by the genotype. Second, other abiotic or biotic selection pressures may be more potent drivers of local adaptation in redbreast sunfish than the environmental variables I measured. For example, a great deal of variation in species assemblages occurs along the basins inhabited by redbreast sunfish and it is possible that predator-prey interactions have driven local adaptation more strongly than physiographic or climate constraints. Third, parts of the genome not sampled may harbor the locally adapted genetic variation. Given the relatively small number of SNPs I started with – a function of restriction-enzyme cut-site frequency, sequencer limitations, and the number of individuals I combined in the 3RAD library – it was

perhaps unlikely to find many anonymous loci that happened to be adaptively significant and associated with measurable landscape features. Yet despite these important caveats, a long history of common-garden experiments has shown much support for local adaptation and recent genomics studies have also shown this trend (Andrews, Good, Miller, Luikart, & Hohenlohe, 2016; Hereford, Associate Editor: Elizabeth, & Editor: Monica, 2009).

Future Directions and Conclusions

Given the relatively deep phylogeographic break I found between southern and northern basins, which span a relatively small fraction of RBS's range, additional phylogenetic/phylogenomic study may be warranted in RBS, as there may be cryptic taxa within the nominal species. For example, Kim et al. (2022) found evidence for multiple species within Longear Sunfish (*Lepomis megalotis*) and described hybridization within the genus which is thought to be common. In fact, the same genetic data gathered here could be repurposed in this context to analyze for the possibility of cryptic species in an RBS complex. In addition, the explicit linking of the environmental variation experienced by RBS to phenotypic variation could present a useful avenue, and one initially planned in this study, to explore as the phenotypic variation is what selection acts on and likely plays a role in RBS adaptability. Achieving this goal would require a deeper exploration of the RBS genome, sequencing perhaps an order of magnitude more RAD loci, in order to increase the chances of detecting loci under strong selection. Going further, whole genomic sequencing (WGS) could be used to create a chromosomal-level assembly in RBS or any other *Lepomis sp.* in order to place markers, such as those found here, in the context of a genomic architecture, and studies of gene expression could show which genes are more or less expressed based on environmental heterogeneity. Finally, as previously mentioned, some fine-grained questions about RBS gene flow remain unresolved, in particular the typical rate of gene flow per unit of stream distance (fine-scale IBD) and how this varies in different ecosystems and ecoregions.

In conclusion, this study aimed to use relatively new genomic sequencing methods to characterize the genetic diversity of a widely distributed, generalist freshwater fish. By comparing neutral genetic variability to geographic and environmental distances in dendritic river systems, this study adds to the

body of knowledge concerning generalities in the way population are structured in heterogeneous river environments. The putatively adaptive loci discovered in this study illustrate potential evidence of local adaptation in wild populations. The use of environmental data as predictor variables to detect these loci represents a new and promising approach that lacks widespread usage and testing, which I have done here, though further characterization of these loci is needed. Overall, this study should provide useful information for evolutionary biologists, conservation biologists, managers, and ecologists, especially those interested in freshwater fish of the southeastern United States.

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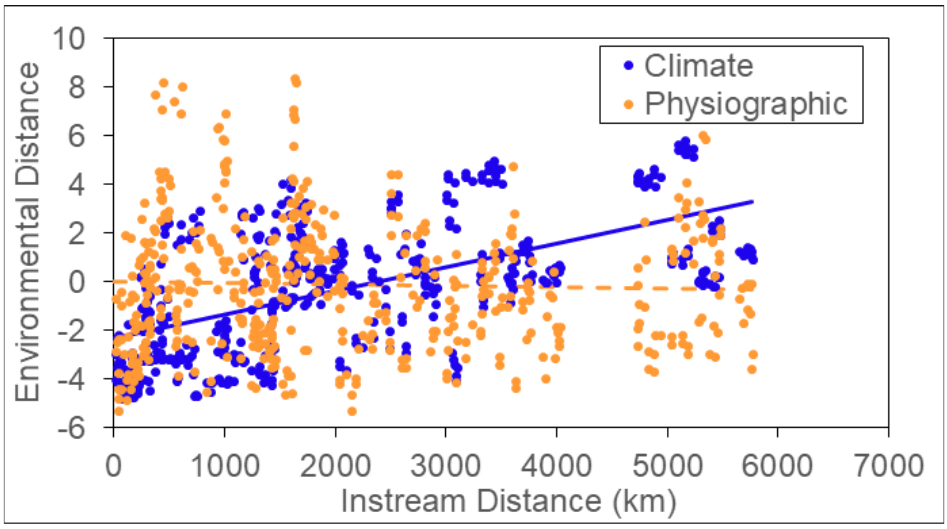
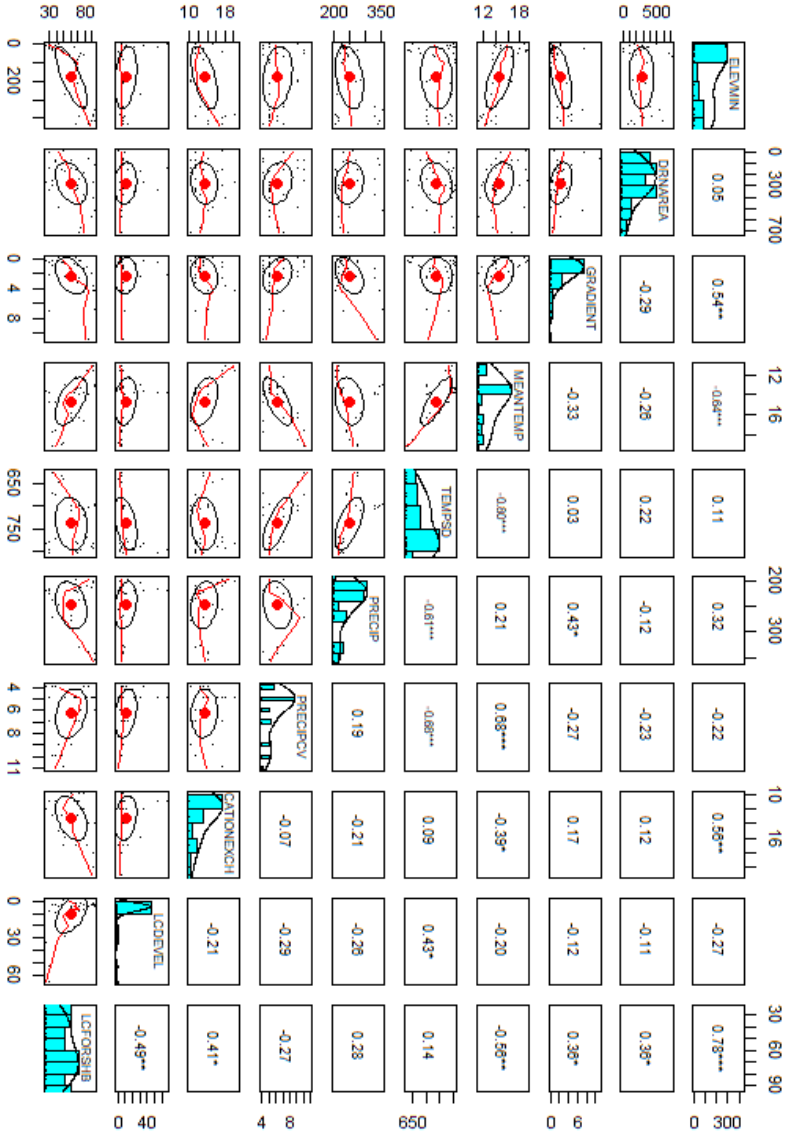
APPENDIX 1

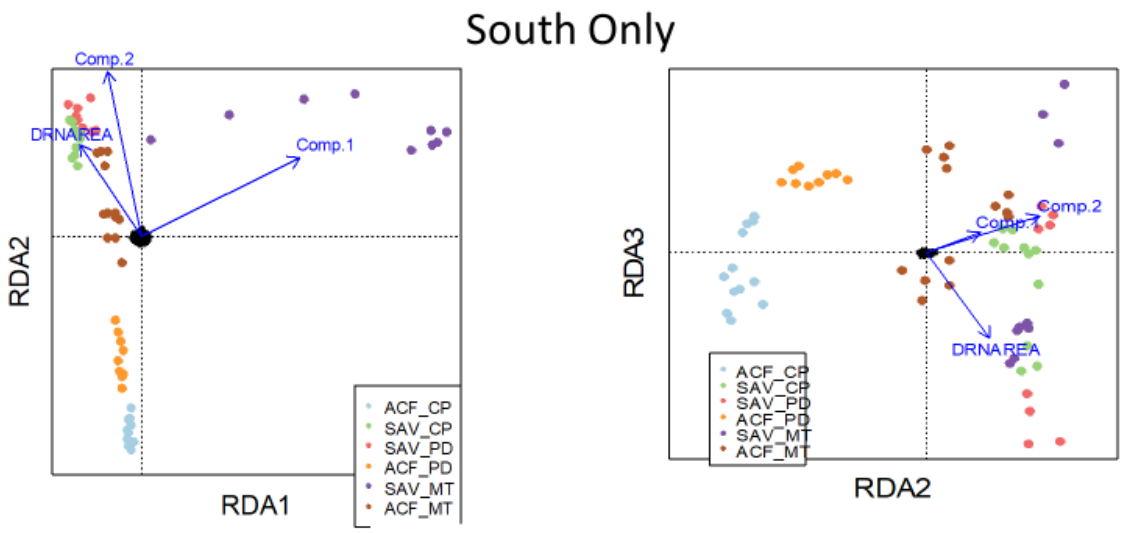
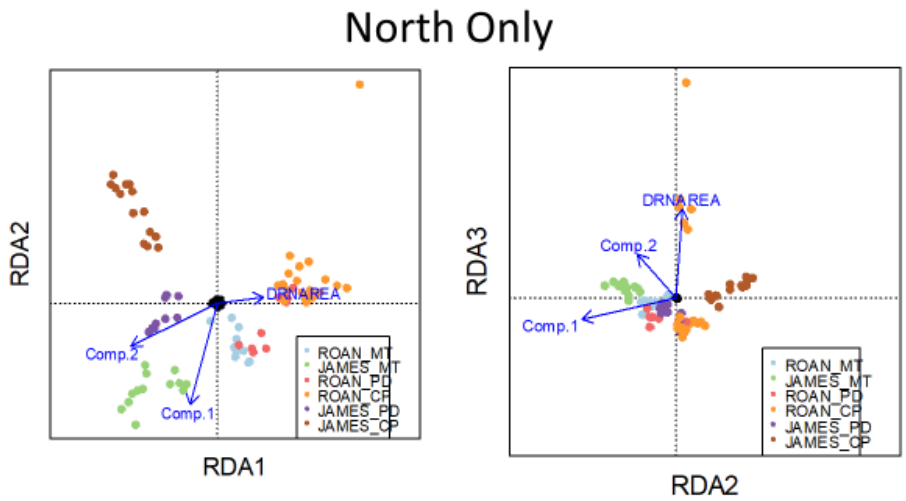
CLIMATE AND PHSIOGRAPHIC COMPOSITE VARIABLES VS. INSTREAM DISTANCE



APPENDIX 2

ENVIRONMENTAL VARIABLE CORRELATIONS



APPENDIX 3

RDA TRIPLOTS FROM BOTH NORTHERN AND SOUTHERN BASINS